

ROADWAY STANDARD DRAWINGS

October 2016

REVISION HISTORY

Ed/Rev Number	Drawing Number	Description of Revision	Authorised By	Date
Ed 1/ Rev 0	All	Review drawings as part of a review of the Engineering Guidelines	Executive	October 2014
Ed 1/ Rev 1	EGSD-303	Amend to bring in line with Landscape Standard Drawings and Construction Guidelines	SDSG	8/02/2016
	Index Page	Include reference to RMS drawings where no LMCC drawings apply		
Ed 1/ Rev 2	All	Reformat and minor changes	SDSG	5/09/2016
	EGSD-202	Add a new non-standard drawing for retrofitting a driveway in steep terrain		
	EGSD-301	Increase the spacing of joints in a concrete footpath		
	EGSD-302	Adopt 1m as the standard height for bollards		
	EGSD-303	Change utility allocation in footway to reflect current practices		
	EGSD-702	Add a new drawing for a minimum road standard		
	EGSD-703	Add a new drawing for the restoration conditions of a trench in the footway		
	EGSD-704	Add new drawings for the restoration of a trench including conditions, in the roadway		

Note: SDSG is the Standard Drawings Steering Group

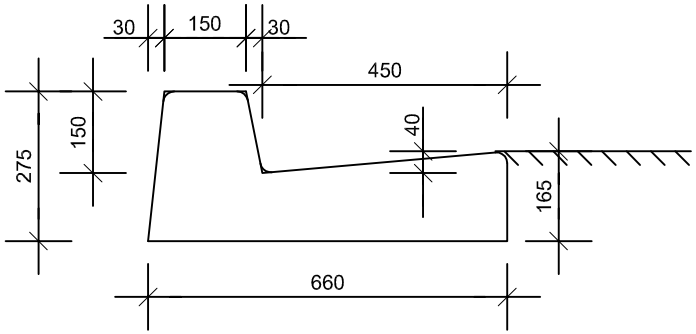
The Standard Drawings complement the engineering and landscape guidelines by providing design detail of infrastructure requirements.

Where the LMCC Standard Drawings do not cover a particular design, the RMS standards included at www.rms.nsw.gov.au/business-industry/partners-suppliers/design-documents/index.html are to apply.

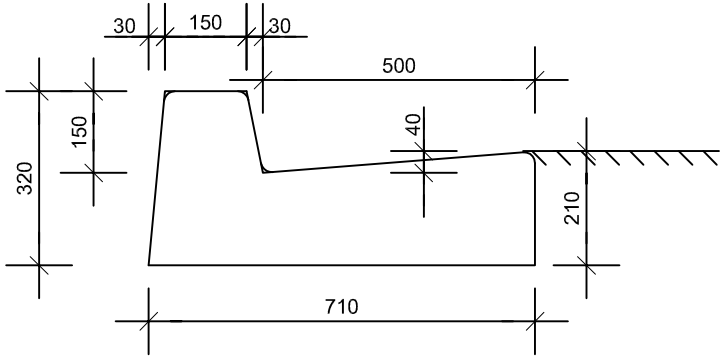
The Roadway Drawings include the following:

CATEGORY	DRAWING NUMBER	DESCRIPTION	NO OF SHEETS
KERB AND GUTTER			
	EGSD-101	KERB PROFILES AND DIMENSIONS	2
	EGSD-102	KERB RAMPS	3
	EGSD-103	VEHICLE CROSSING – RESIDENTIAL	2
	EGSD-104	COMMERCIAL AND INDUSTRIAL VEHICLE CROSSING	1
DRIVEWAYS			
	EGSD-201	RESIDENTIAL DRIVEWAY	2
	EGSD-202	DRIVEWAY PROFILES	3
	EGSD-203	RURAL DRIVEWAYS – NO KERB & GUTTER	1
FOOTPATHS			
	EGSD-301	CONCRETE FOOT PAVING	3
	EGSD-302	PATHWAY BARRIERS – BOLLARDS	1
	EGSD-303	FOOTWAY ALLOCATION UTILITY SERVICES AND TREES	1
	EGSD-304	FOOTPATH CHICANE	1
MISCELLANEOUS			
	EGSD-701	TURNING AREA FOR CUL-DE-SACS	1
	EGSD-702	PROVISIONAL AND MINIMUM ROAD STANDARDS	1
	EGSD-703	SERVICE TRENCH FOOTWAY RESTORATION	1
	EGSD-704	SERVICE TRENCH ROAD RESTORATION UP TO 1.2M WIDTH	2

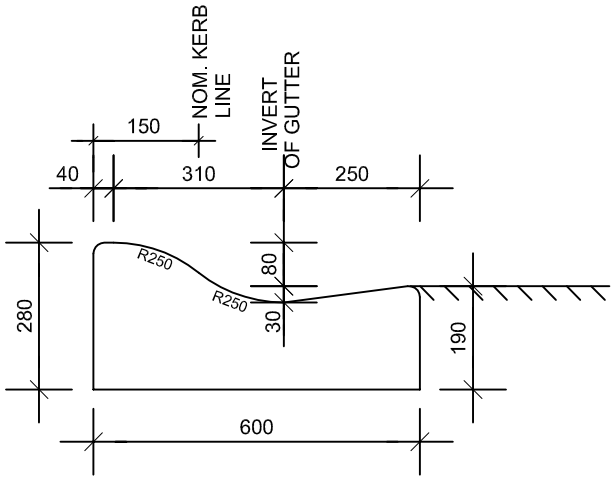
STANDARD KERB & GUTTER
(LMCC)



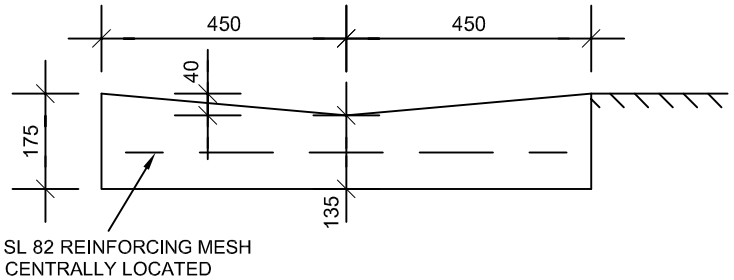
SA
(RTA)



ROLL TOP KERB
(MODIFIED RT)

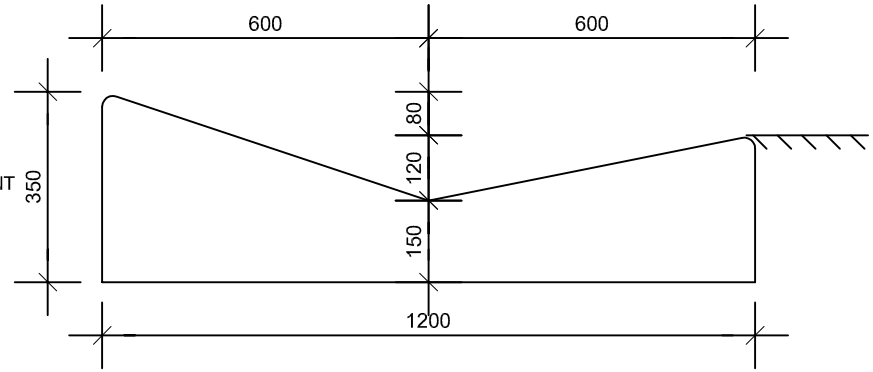


SB
DISHED CROSSING



SL 82 REINFORCING MESH
CENTRALLY LOCATED

SK
GUTTER ADJACENT TO SHOULDERS IN CUTTING



NOTES:

1. ALL EXPOSED EDGES TO BE ROUNDED TO 20mm RADIUS
2. ALL KERBS TO BE CONSTRUCTED WITH 25MPa CONCRETE
3. CONTROL JOINTS TO BE PROVIDED EVERY 2.5-3m
4. EXPANSION JOINTS EVERY 15m AND AT EVERY LAYBACK, LINTEL AND PRAM RAMP
5. ADMIXTURE TO BE USED FOR FINISHING SURFACES
6. ALL DIMENSIONS ARE IN MILLIMETRES

VERSION	DATE	COMMENTS	PAPER SIZE
04	29/07/16	AMENDMENT	A4
03	29/05/13	AMENDMENT	A4
02	04/07/12	AMENDMENT	A4
01	18/10/11	ORIGINAL ISSUE	A4

WORK TO FIGURED DIMENSIONS - DO NOT SCALE. DO NOT RELY ON THESE STANDARD DRAWINGS AS THE EQUIVALENT OF, OR SUBSTITUTE FOR, PROJECT SPECIFIC DESIGN AND ASSESSMENT BY A QUALIFIED PROFESSIONAL.

Lake Macquarie City Council
126-138 MAIN ROAD
SPEERS POINT
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DRAWING TITLE: KERB PROFILES & DIMENSIONS

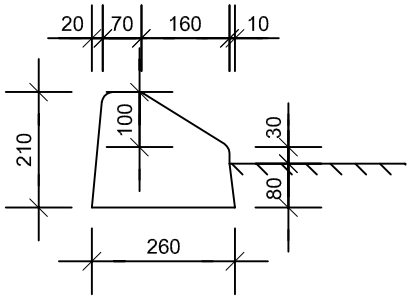
SHEET: 1 OF 2

DRAWING No. EGSD-101

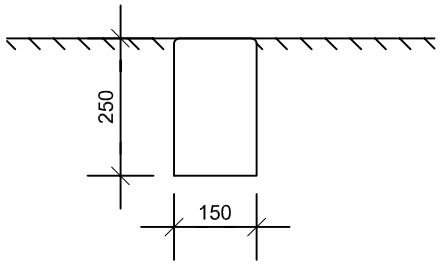
VERSION: 04

SF

RAISED MEDIANS
& TRAFFIC ISLANDS

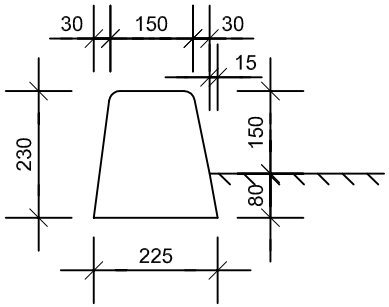


CONCRETE EDGE



SM

BARRIER KERB
AT TRAFFIC ISLANDS



NOTES:

1. ALL EXPOSED EDGES TO BE ROUNDED TO 20mm RADIUS
2. ALL KERBS TO BE CONSTRUCTED WITH 25MPa CONCRETE
3. ALL DIMENSIONS ARE IN MILLIMETRES

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01	18/10/11	ORIGINAL ISSUE	A4

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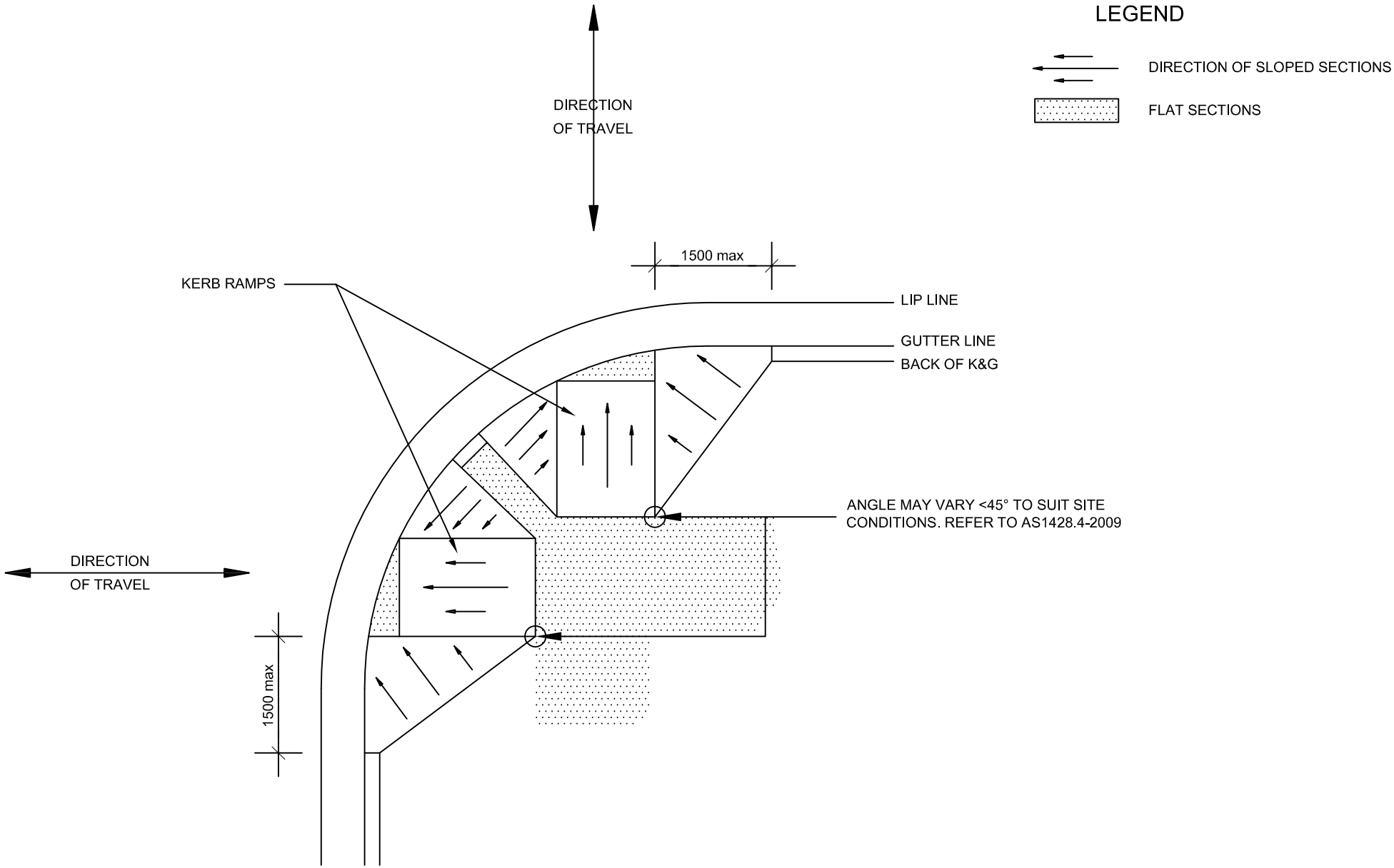
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DRAWING TITLE
KERB PROFILES & DIMENSIONS

SHEET: 2 OF 2

DRAWING No. EGSD-101

VERSION: 02



TYPICAL LOCATION OF KERB RAMPS

VERSION	DATE	COMMENTS	PAPER SIZE
02	17/06/15	AMENDMENT	A4
01	18/10/11	ORIGINAL ISSUE	A4

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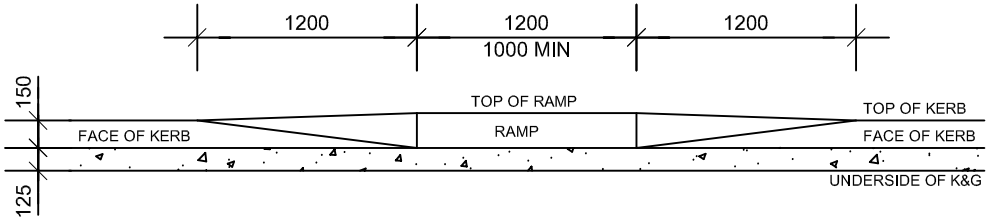
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DRAWING TITLE: KERB RAMPS LAYOUT

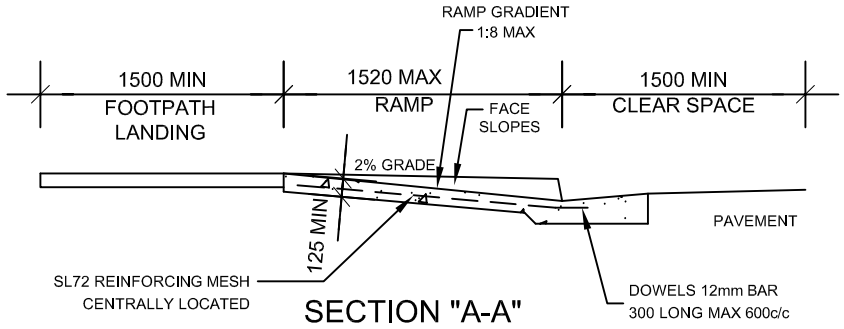
SHEET: 1 OF 3

DRAWING No. EGSD-102

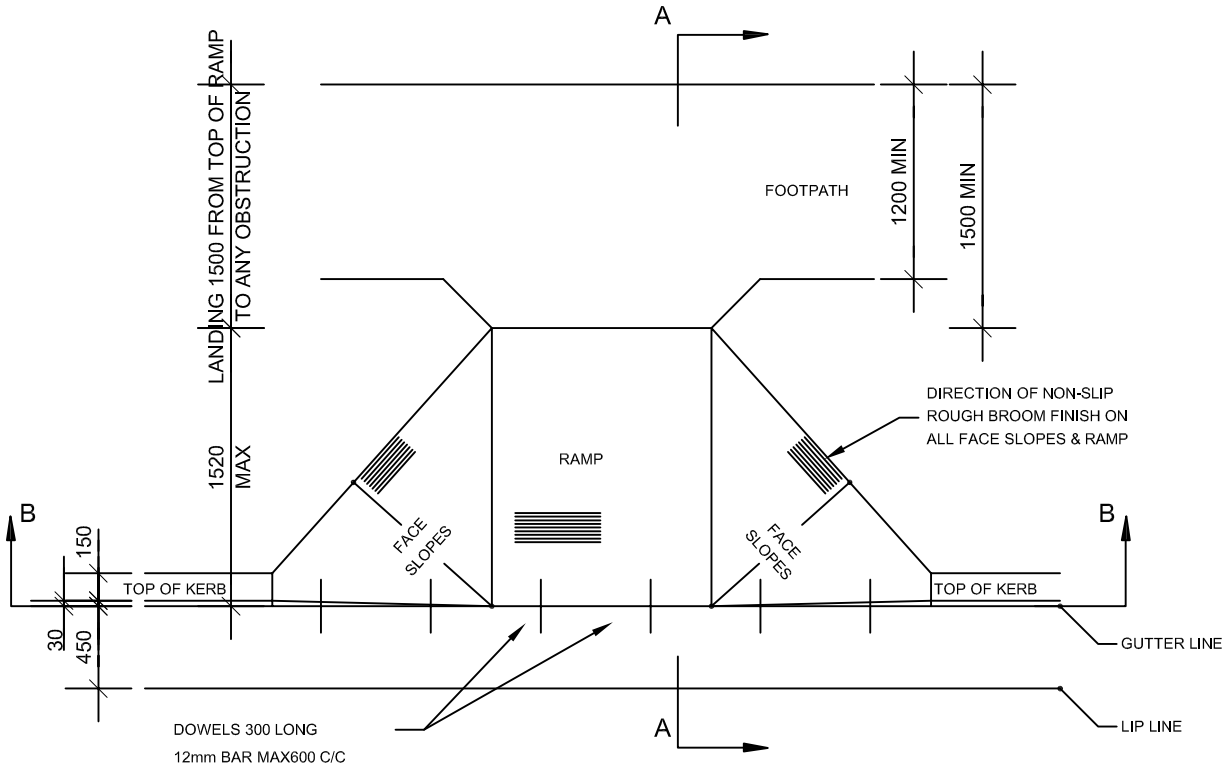
VERSION: 02



SECTION "B-B"



SECTION "A-A"



PLAN VIEW

NOTES

1. RAMPS TO FACE DIRECTION OF TRAVEL
2. THIS DRAWING IS AN EXAMPLE ONLY "BUT" IS INDICATIVE OF THE STANDARD REQUIREMENTS
3. THE DESIGN OF EACH RAMP WILL BE SITE SPECIFIC TO SUIT LOCAL CONDITIONS
4. KERB RAMP TO BE CAST IN ONE POUR, WITH NO JOINTS WITHIN THE RAMP. NO CONSTRUCTION JOINTS WILL BE ALLOWED
5. DESIGNED TO AS 1428.1-2009 AND 1428.4-2009 RAMP COLOUR TO BE "SANDSTONE". IT MAY BE CHANGED WITH COUNCIL APPROVAL BUT ONLY TO CONTRAST WITH ADJOINING SURFACES
6. FOR KERB RAMPS ABUTTING A CROSSING NOT SQUARE TO KERB LINE REFER TO AS 1424.4-2009
7. RAMP TO BE CONSTRUCTED WITH 25MPa CONCRETE
8. TACTILE GROUND SURFACE INDICATORS TO BE PLACED WHERE REQUIRED. FOR FURTHER DETAILS SEE AS 1428.4-2009
9. ALL DIMENSIONS ARE IN MILLIMETRES

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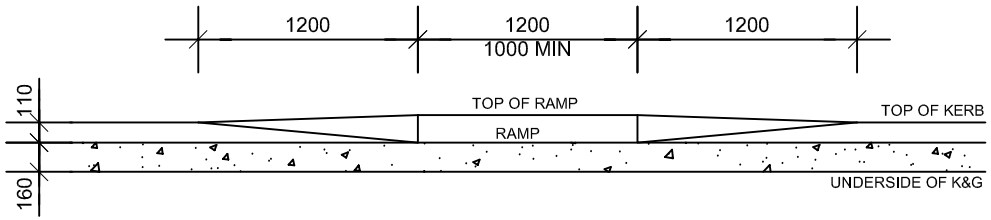
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126-138 MAIN ROAD
SPEERS POINT
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DRAWING TITLE
KERB RAMPS
FOR STANDARD KERB & GUTTER

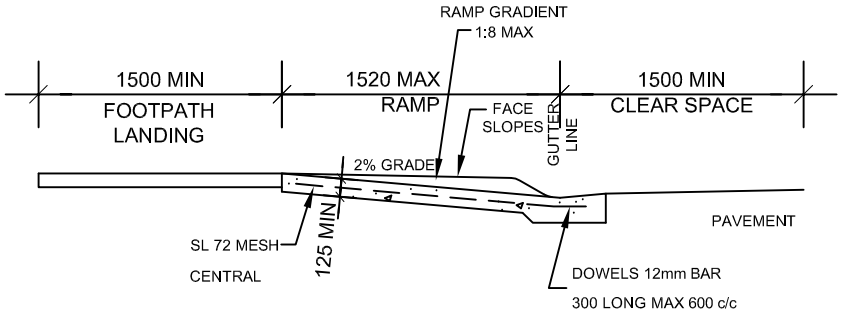
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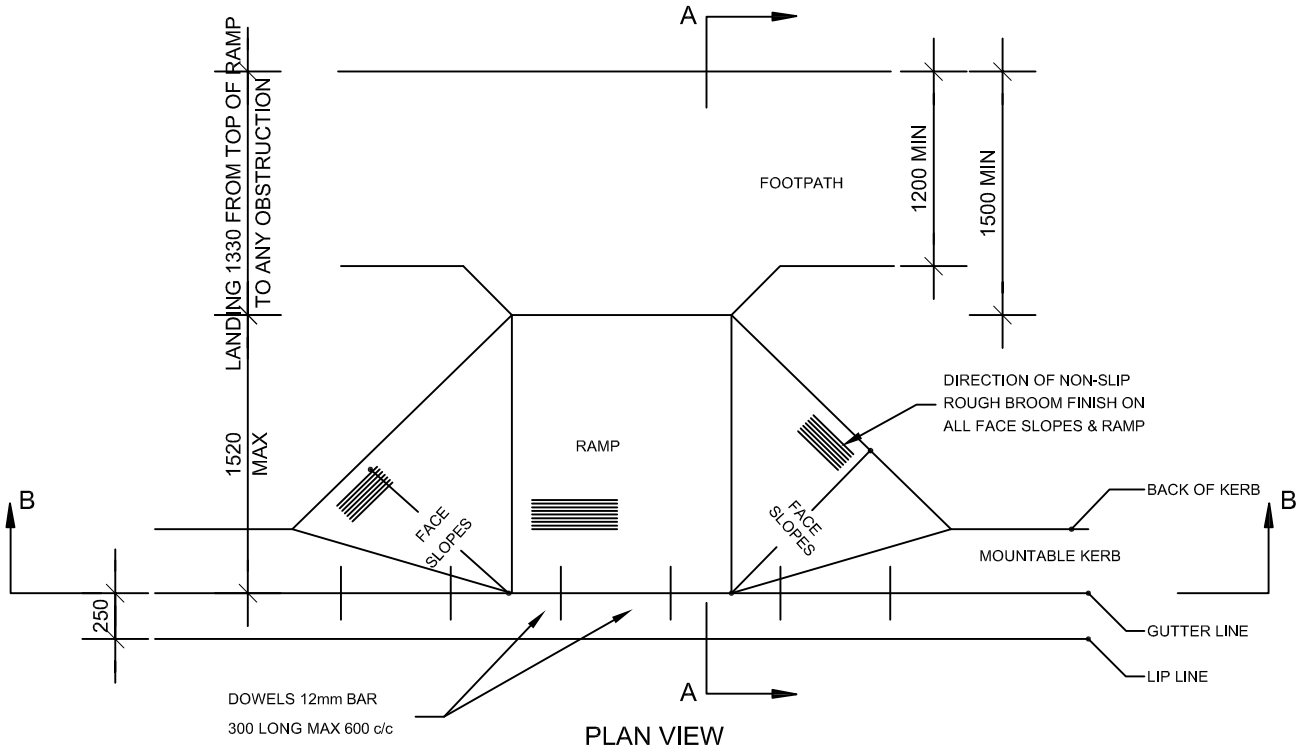
VERSION: 04



SECTION "B-B"



SECTION "A-A"



PLAN VIEW

NOTES

1. RAMPS TO FACE DIRECTION OF TRAVEL
2. THIS DRAWING IS AN EXAMPLE ONLY "BUT" IS INDICATIVE OF THE STANDARD REQUIREMENTS
3. THE DESIGN OF EACH RAMP WILL BE SITE SPECIFIC TO SUIT LOCAL CONDITIONS
4. KERB RAMP TO BE CAST IN ONE POUR, WITH NO JOINTS WITHIN THE RAMP. NO CONSTRUCTION JOINTS WILL BE ALLOWED
5. DESIGNED TO AS 1428.1-2009 AND 1428.4-2009 RAMP COLOUR TO BE "SANDSTONE". IT MAY BE CHANGED WITH COUNCIL APPROVAL BUT ONLY TO CONTRAST WITH ADJOINING SURFACES
6. FOR KERB RAMPS ABUTTING A CROSSING NOT SQUARE TO KERB LINE REFER TO AS 1424.4-2009
7. RAMP TO BE CONSTRUCTED WITH 25MPa CONCRETE
8. TACTILE GROUND SURFACE INDICATORS TO BE PLACED WHERE REQUIRED. FOR FURTHER DETAILS SEE AS 1428.4-2009
9. ALL DIMENSIONS ARE IN MILLIMETRES

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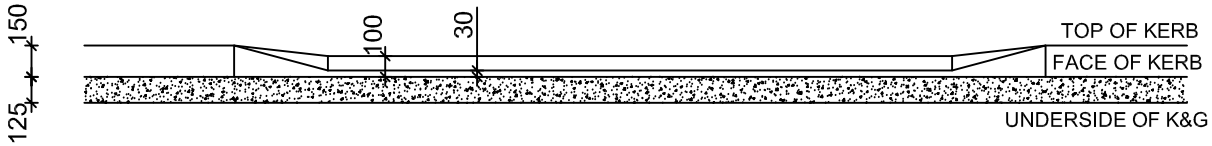
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DRAWING TITLE
KERB RAMPS
FOR ROLL TOP KERB

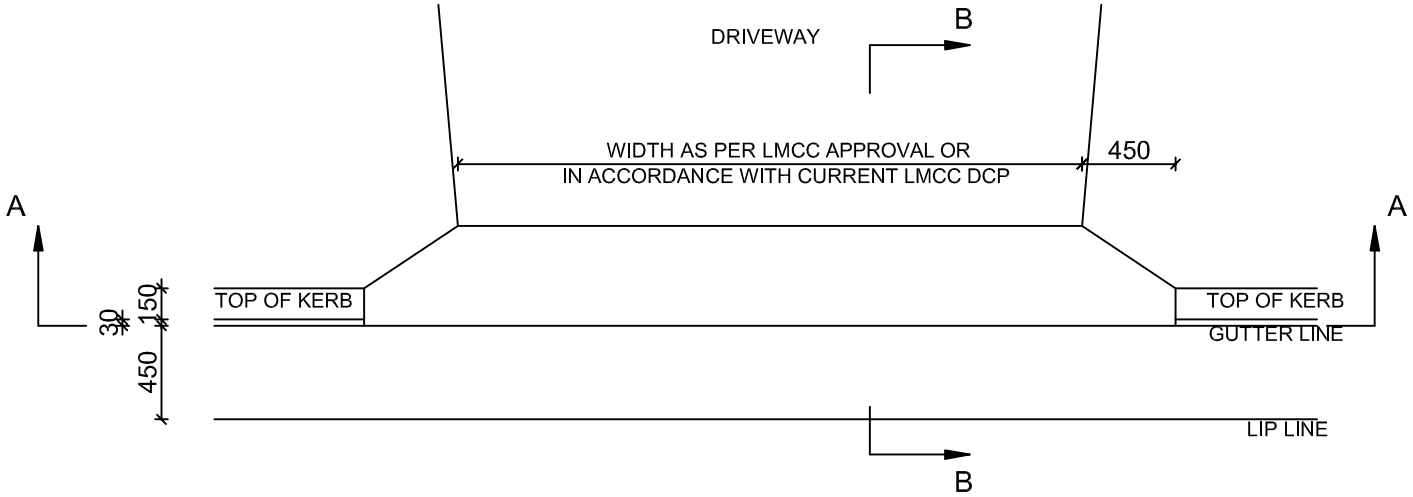
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DRAWING No. EGSD - 102

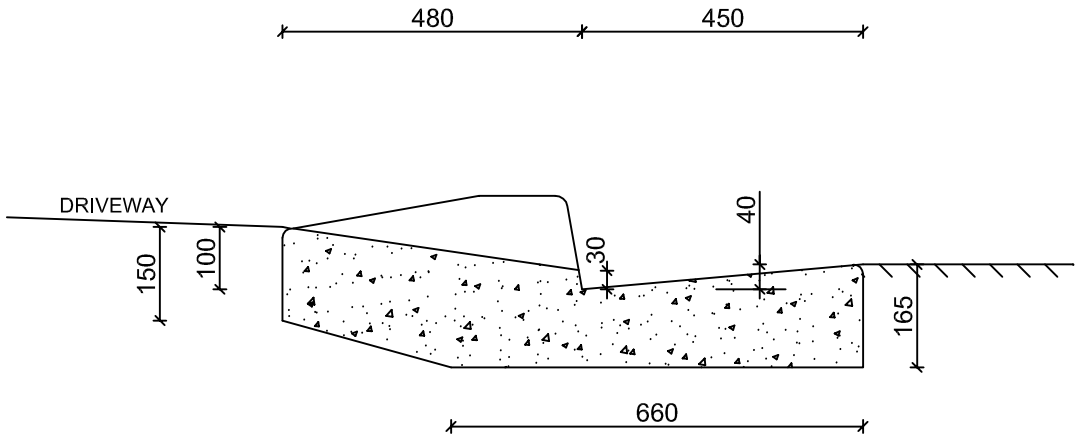
VERSION: 04



SECTION "A-A"



PLAN VIEW



SECTION "B-B"

NOTES

1. RESIDENTIAL CROSSING TO BE CONSTRUCTED WITH 25MPa CONCRETE
2. WEAKENED PLANE CONTROL JOINTS IN MACHINE MOULDED K&G OR EXPANSION JOINTS IN HAND CONSTRUCTED K&G TO BE PROVIDED AT EACH END OF VEHICULAR CROSSING
3. AT LARGER VEHICLE CROSSING A TOOL FORMED JOINT IS TO BE CONSTRUCTED IN THE CROSSING AT MAX 4.0m SPACINGS
4. FOR OTHER DRIVEWAY WIDTHS REFER TO CURRENT DCP
5. FOR CONCRETE DRIVEWAY, DOWEL INTO CONCRETE AS PER EGSD-201
6. WHERE THE DRIVEWAY CROSSES AN EXISTING GRATED GULLY PIT, A QUALIFIED ENGINEER IS TO CERTIFY THAT THERE IS NO LOSS OF CAPACITY IN THE DRAINAGE STRUCTURE
7. ALL DIMENSIONS ARE IN MILLIMETRES

VERSION	DATE	COMMENTS	PAPER SIZE
04	29/07/16	AMENDMENT	A4
03	11/12/13	AMENDMENT	A4
02	29/05/13	AMENDMENT	A4
01	18/10/11	ORIGINAL ISSUE	A4

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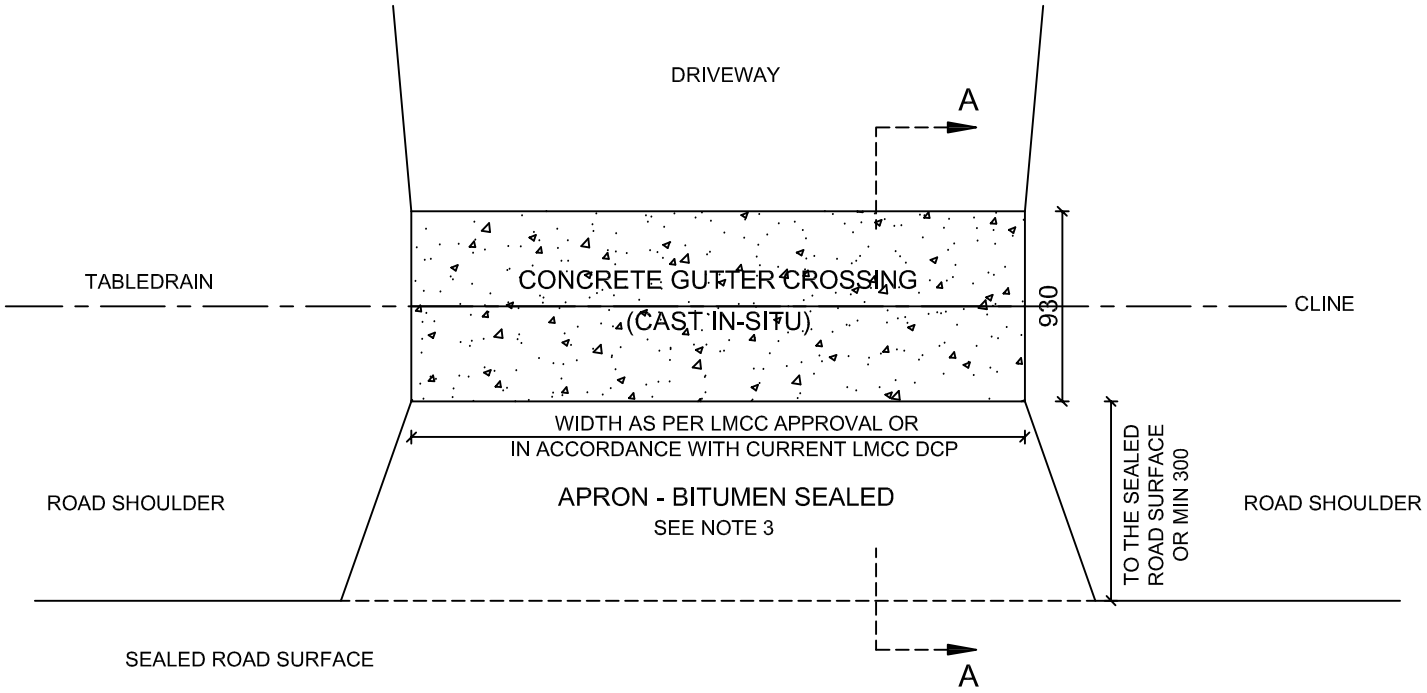
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DRAWING TITLE
VEHICLE CROSSING - RESIDENTIAL
STANDARD K&G

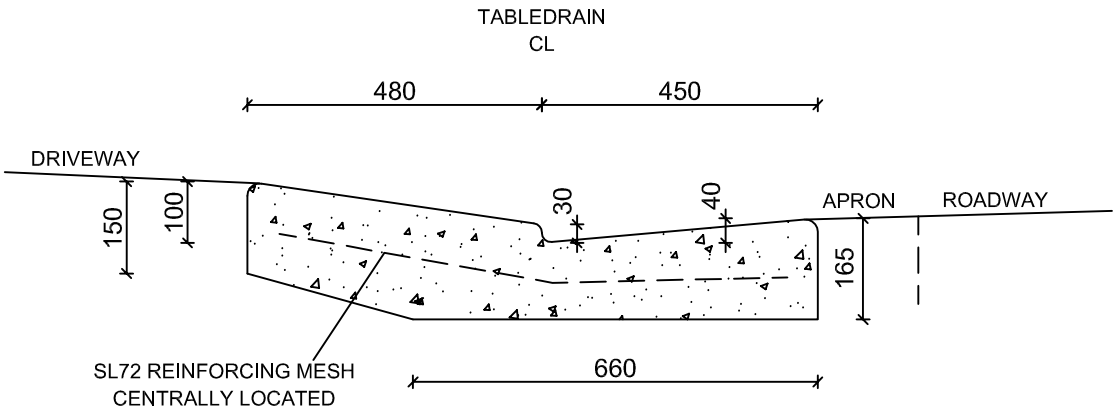
SHEET: 1 OF 2

DRAWING No. EGSD - 103

VERSION: 04



PLAN VIEW



SECTION "A-A"

NOTES

1. GUTTER CROSSING TO BE CONSTRUCTED WITH 25MPa CONCRETE
2. FOR CONCRETE DRIVEWAY, DOWEL INTO CONCRETE AS PER EGSD-201 ALL
3. APRON TO BE SEALED BY PROPERTY OWNER THEN MAINTAINED BY COUNCIL
4. ALL DIMENSIONS ARE IN MILLIMETRES

VERSION	DATE	COMMENTS	PAPER SIZE
03	29/07/16	AMENDMENT	A4
02	29/05/13	AMENDMENT	A4
01	18/10/11	ORIGINAL ISSUE	A4

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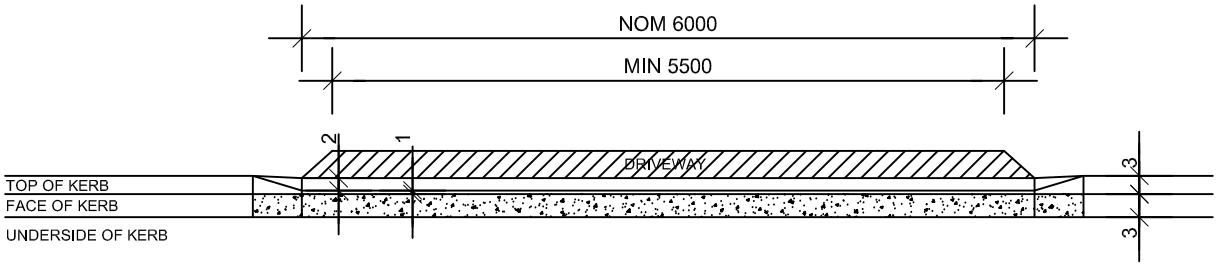
Lake Macquarie City Council
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PHONE (02) 4921 0333
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DRAWING TITLE
VEHICLE CROSSING - RESIDENTIAL
GUTTER CROSSING WHERE NO K&G

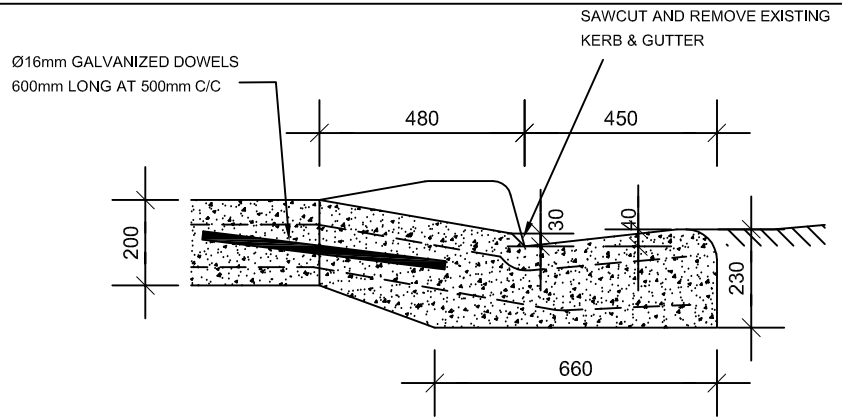
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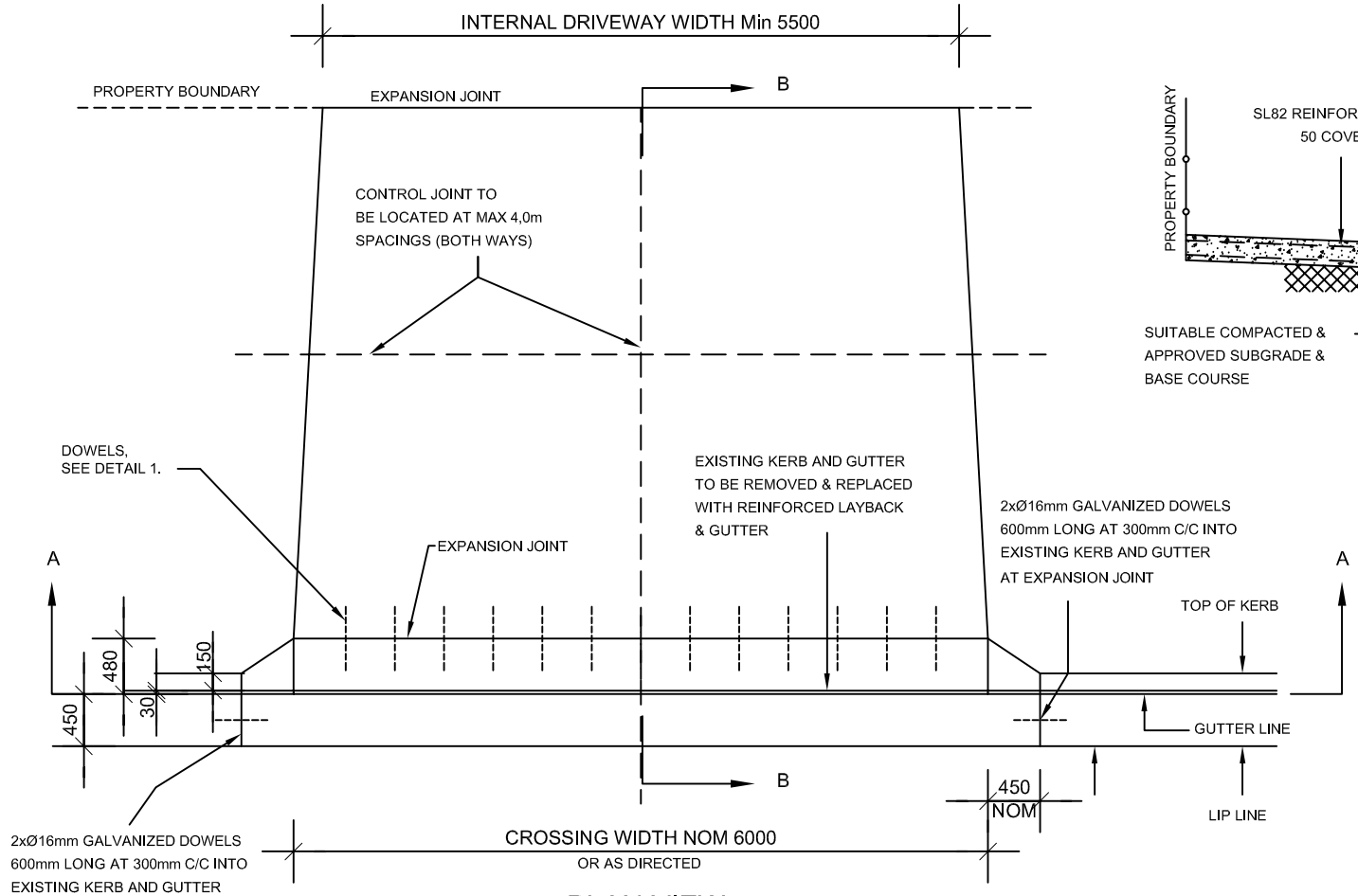
VERSION: 03



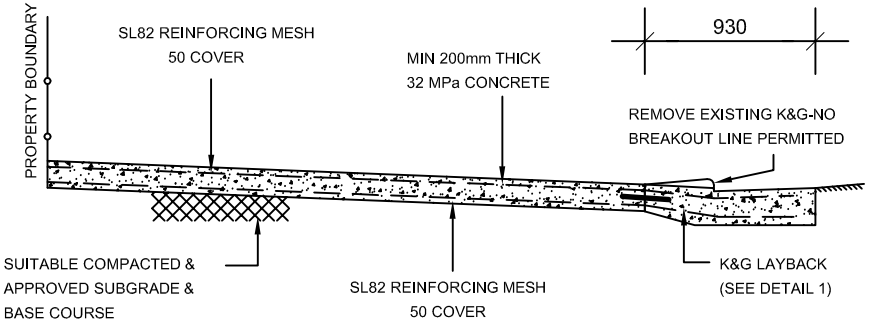
SECTION "A-A"



DETAIL 1



PLAN VIEW



SECTION "B-B"

NOTES

1. WHERE SUBGRADE CONDITIONS ARE POOR AND/OR THE LOADINGS OF THE CROSSING MAY BE EXCESSIVE, THE CROSSING IS TO BE DESIGNED BY A QUALIFIED ENGINEER
2. FOR CROSSING & DRIVEWAY WIDTHS REFER TO CURRENT DCP
3. WHERE THE DRIVEWAY CROSSES AN EXISTING GRATED GULLY PIT, A QUALIFIED ENGINEER IS TO CERTIFY THAT THERE IS NO LOSS OF CAPACITY IN THE DRAINAGE STRUCTURE
4. ALL DIMENSIONS ARE IN MILLIMETRES

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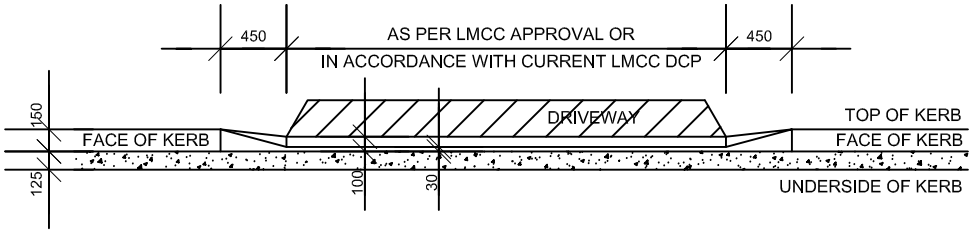
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DRAWING TITLE
COMMERCIAL AND INDUSTRIAL VEHICLE DRIVEWAY & CROSSING

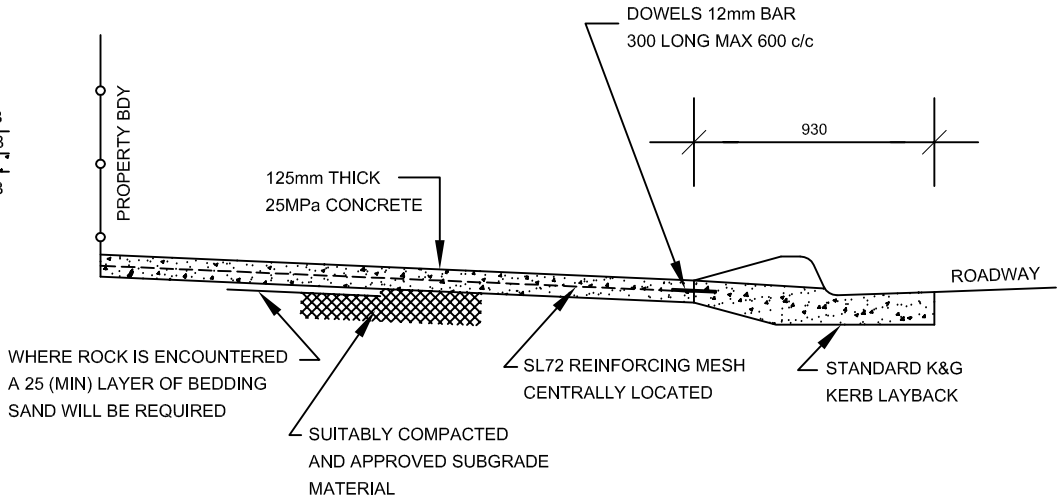
SHEET: 1 OF 1

DRAWING No. EGSD - 104

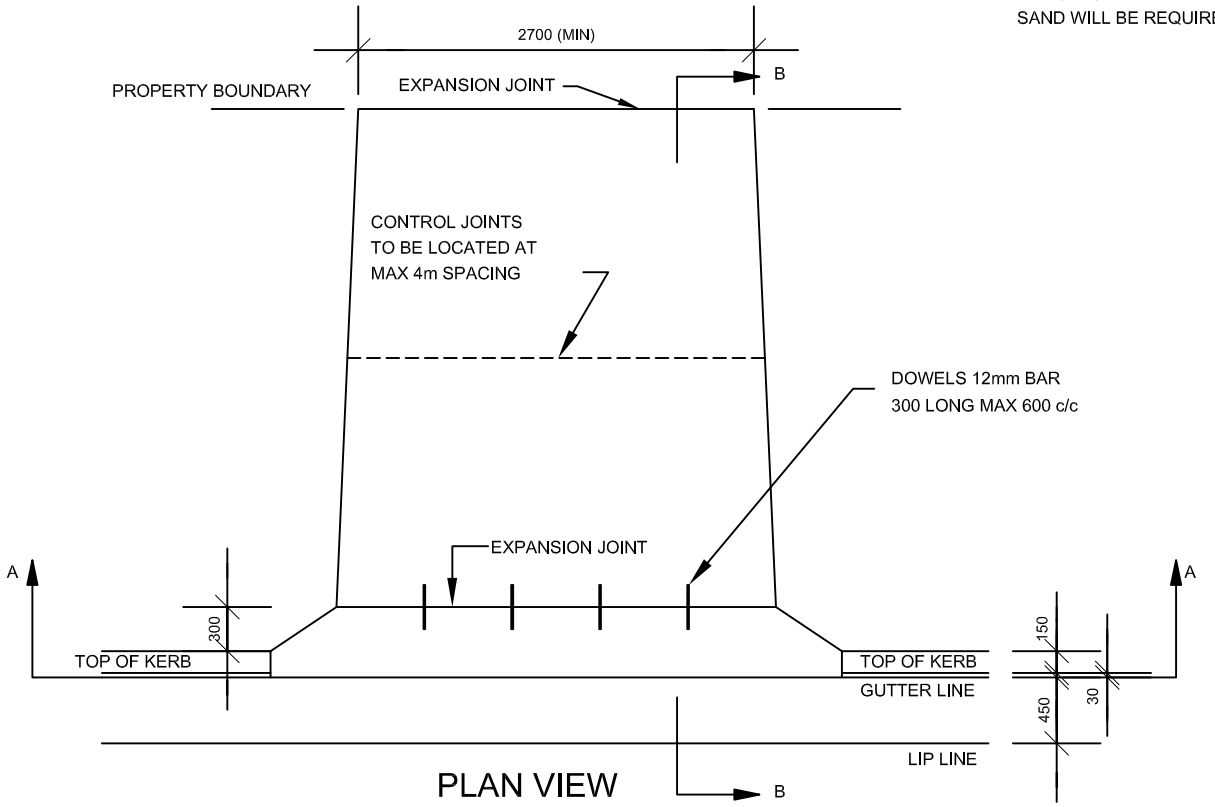
VERSION: 04



SECTION "A-A"



SECTION "B-B"



PLAN VIEW

NOTES

1. FOR DRIVEWAY WIDTHS REFER TO CURRENT DCP
2. REFER TO EGSD-103 FOR DETAILS OF KERB CROSSING
3. ALL DIMENSIONS ARE IN MILLIMETRES

VERSION	DATE	COMMENTS	PAPER SIZE
03	29/07/16	AMENDMENT	A4
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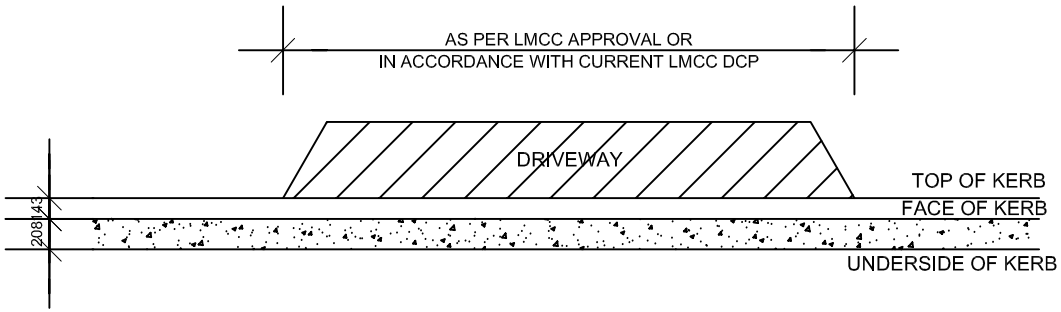
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DRAWING TITLE
RESIDENTIAL DRIVEWAY
STANDARD KERB AND GUTTER

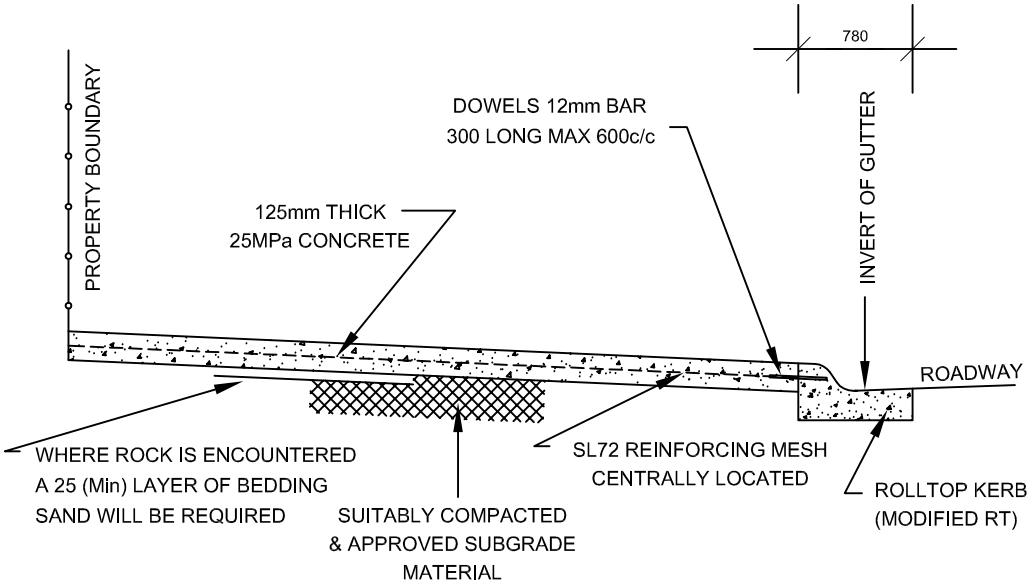
SHEET: 1 OF 2

DRAWING No. EGSD - 201

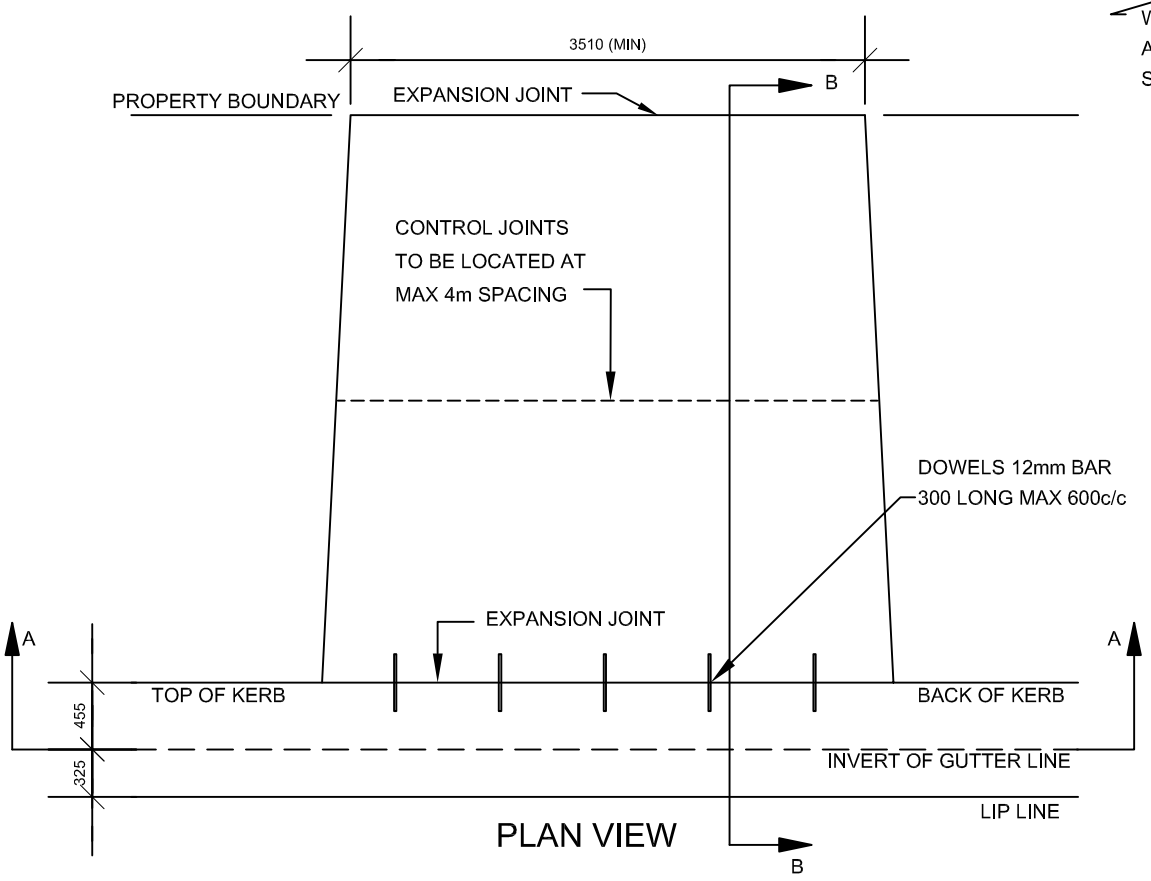
VERSION: 03



SECTION "A-A"



SECTION "B-B"



PLAN VIEW

NOTES

1. FOR DRIVEWAY WIDTHS REFER TO CURRENT DCP
2. REFER TO EGSD-103 FOR DETAILS OF KERB CROSSING
3. ALL DIMENSIONS ARE IN MILLIMETRES

VERSION	DATE	COMMENTS	PAPER SIZE
03	29/07/16	AMENDMENT	A4
02	29/05/13	AMENDMENT	A4
01	18/10/11	ORIGINAL ISSUE	A4

WORK TO FIGURED DIMENSIONS - DO NOT SCALE. DO NOT RELY ON THESE STANDARD DRAWINGS AS THE EQUIVALENT OF, OR SUBSTITUTE FOR, PROJECT SPECIFIC DESIGN AND ASSESSMENT BY A QUALIFIED PROFESSIONAL.

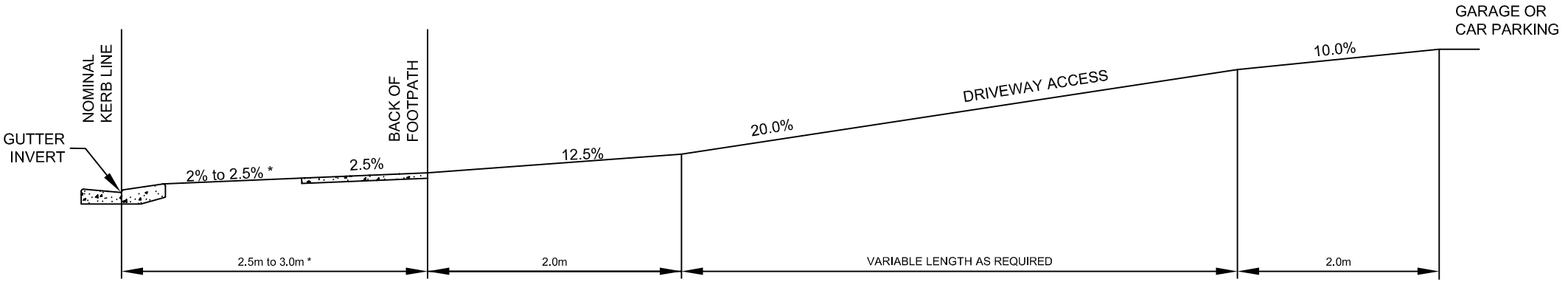
Lake Macquarie City Council
126-138 MAIN ROAD
SPEERS POINT
PHONE (02) 4921 0333
www.lakemac.com.au

DRAWING TITLE
RESIDENTIAL DRIVEWAY
ROLL TOP KERB

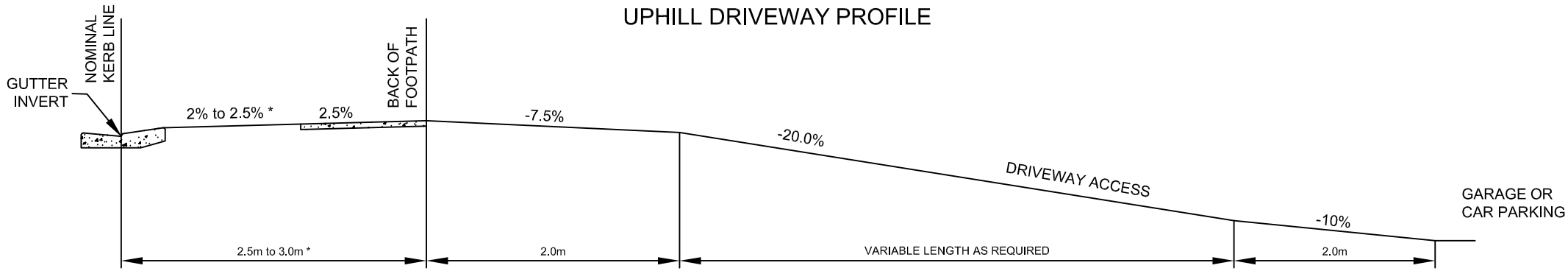
SHEET: 2 OF 2

DRAWING No. EGSD - 201

VERSION: 03



UPHILL DRIVEWAY PROFILE



DOWNHILL DRIVEWAY PROFILE

* depending on footway width (EGSD-303)

NOTES:

1. GRADING AND DIMENSIONS SHOWN SATISFY REQUIREMENTS OF AS/NZS 2890.1:2004 FOR B85 VEHICLES.
2. THE RECOMMENDED MAXIMUM CHANGE IN GRADE ACROSS A HINGE POINT SHOULD BE LIMITED TO 12.5% FOR A CONVEX TRANSITION OR 15% FOR A CONCAVE TRANSITION.
3. THE MAXIMUM GRADIENT OF A DOMESTIC DRIVE WAY WITHIN THE PROPERTY SHALL BE 1 in 5 (20%) IN ACCORDANCE WITH CURRENT DCP. WHERE EXISTING TOPOGRAPHY EXCEEDS 1 IN 5, THEN A MAXIMUM GRADE OF 1 IN 4 (25%) MAY BE USED WITH COUNCIL APPROVAL.
4. A 2.0m TRANSITION IS REQUIRED TO ANY PARKING AREA OR GARAGE AT A GRADE OF 10%.
5. WHERE AN ACCESS DRIVEWAY IS PROPOSED WITHIN AN OLDER DEVELOPED AREA AND WHERE A LARGE DIFFERENCE IN LEVEL BETWEEN THE EXISTING ROADWAY AND THE PROPERTY BOUNDARY OCCURS, THEN A SITE SPECIFIC DESIGN MAY BE REQUIRED WHICH MAY INCLUDE LOWERING THE EXISTING FOOTPATH. THESE ARE NON-STANDARD TREATMENTS AND WILL REQUIRE DESIGN BY A SUITABLY QUALIFIED PROFESSIONAL.
6. THE GRADES INDICATED WILL PROVIDE SATISFACTORY VEHICULAR ACCESS FOR MOST STANDARD VEHICLES. PROPERTY OWNERS SHOULD BE AWARE THAT SOME DIFFICULTIES WITH ACCESS MAY STILL OCCUR WITH MODIFIED VEHICLES, EXOTIC SPORTS CARS, AND TRAILERS INCLUDING CARAVANS AND BOAT TRAILERS.
7. ANY PROFILE EXCEEDING ANY OF THESE MAXIMUM SLOPES WILL REQUIRE COUNCIL APPROVAL.

VERSION	DATE	COMMENTS	PAPER SIZE
04	29/07/16	AMENDMENT	A4
03	12/02/15	AMENDMENT	A4
02	29/05/13	AMENDMENT	A4
01	18/10/11	ORIGINAL ISSUE	A4

WORK TO FIGURED DIMENSIONS - DO NOT SCALE. DO NOT RELY ON THESE STANDARD DRAWINGS AS THE EQUIVALENT OF, OR SUBSTITUTE FOR, PROJECT-SPECIFIC DESIGN & ASSESSMENT BY A QUALIFIED PROFESSIONAL.

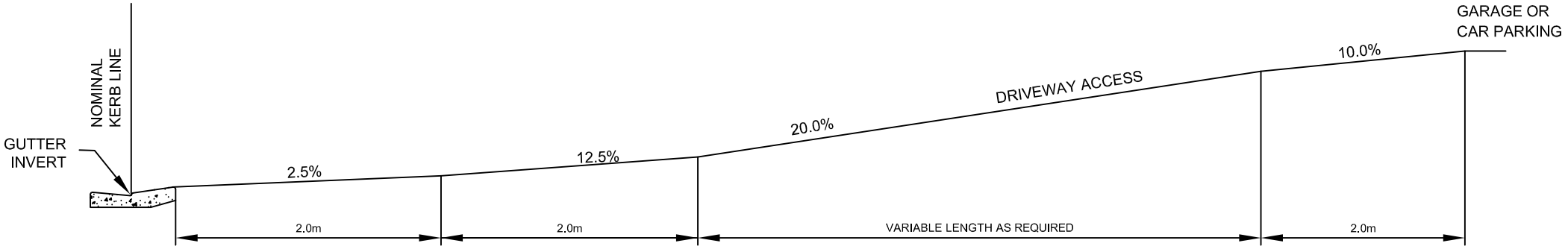
Lake Macquarie City Council
126-138 MAIN ROAD
SPEERS POINT
PHONE (02)4921 0333
www.lakemac.com.au

DRAWING TITLE: DRIVEWAY PROFILES - MAXIMUM GRADES WITH FOOTPATH

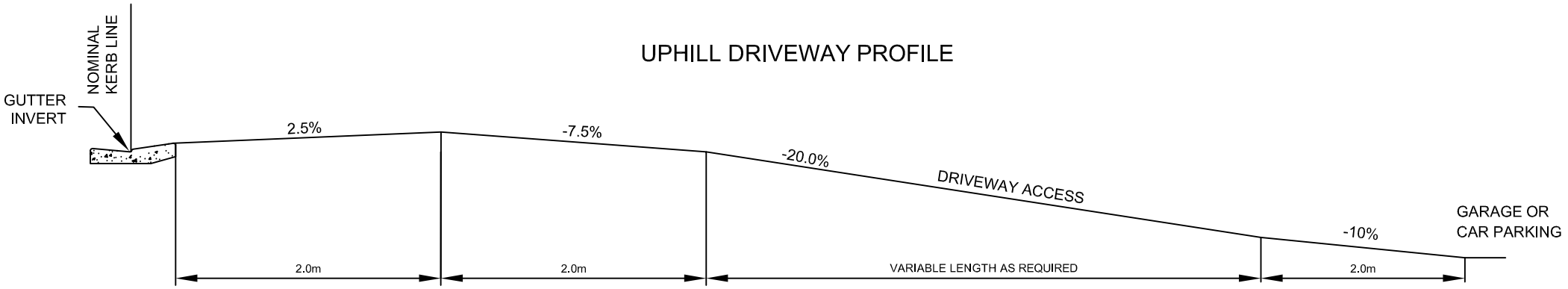
SHEET: 1 OF 3

DRAWING No. EGSD - 202

VERSION: 04



UPHILL DRIVEWAY PROFILE



DOWNHILL DRIVEWAY PROFILE

NOTES:

1. GRADING AND DIMENSIONS SHOWN SATISFY REQUIREMENTS OF AS/NZS 2890.1:2004 FOR B85 VEHICLES.
2. THE RECOMMENDED MAXIMUM CHANGE IN GRADE ACROSS A HINGE POINT SHOULD BE LIMITED TO 12.5% FOR A CONVEX TRANSITION OR 15% FOR A CONCAVE TRANSITION.
3. THE MAXIMUM GRADIENT OF A DOMESTIC DRIVE WAY WITHIN THE PROPERTY SHALL BE 1 in 5 (20%) IN ACCORDANCE WITH CURRENT DCP. WHERE EXISTING TOPOGRAPHY EXCEEDS 1 IN 5, THEN A MAXIMUM GRADE OF 1 IN 4 (25%) MAY BE USED WITH COUNCIL APPROVAL.
4. A 2.0m TRANSITION IS REQUIRED TO ANY PARKING AREA OR GARAGE AT A GRADE OF 10%.
5. WHERE AN ACCESS DRIVEWAY IS PROPOSED WITHIN AN OLDER DEVELOPED AREA AND WHERE A LARGE DIFFERENCE IN LEVEL BETWEEN THE EXISTING ROADWAY AND THE PROPERTY BOUNDARY OCCURS, THEN A SITE SPECIFIC DESIGN MAY BE REQUIRED WHICH MAY INCLUDE LOWERING THE EXISTING FOOTPATH. THESE ARE NON-STANDARD TREATMENTS AND WILL REQUIRE DESIGN BY A SUITABLY QUALIFIED PROFESSIONAL.
6. THE GRADES INDICATED WILL PROVIDE SATISFACTORY VEHICULAR ACCESS FOR MOST STANDARD VEHICLES. PROPERTY OWNERS SHOULD BE AWARE THAT SOME DIFFICULTIES WITH ACCESS MAY STILL OCCUR WITH MODIFIED VEHICLES, EXOTIC SPORTS CARS, AND TRAILERS INCLUDING CARAVANS AND BOAT TRAILERS.
7. ANY PROFILE EXCEEDING ANY OF THESE MAXIMUM SLOPES WILL REQUIRE COUNCIL APPROVAL.

VERSION	DATE	COMMENTS	PAPER SIZE
04	29/07/16	AMENDMENT	
03	17/06/15	AMENDMENT	A4
02	12/05/15	AMENDMENT	A4
01	29/05/13	ORIGINAL ISSUE	A4

WORK TO FIGURED DIMENSIONS - DO NOT SCALE. DO NOT RELY ON THESE STANDARD DRAWINGS AS THE EQUIVALENT OF, OR SUBSTITUTE FOR, PROJECT-SPECIFIC DESIGN & ASSESSMENT BY A QUALIFIED PROFESSIONAL.

Lake Macquarie City Council

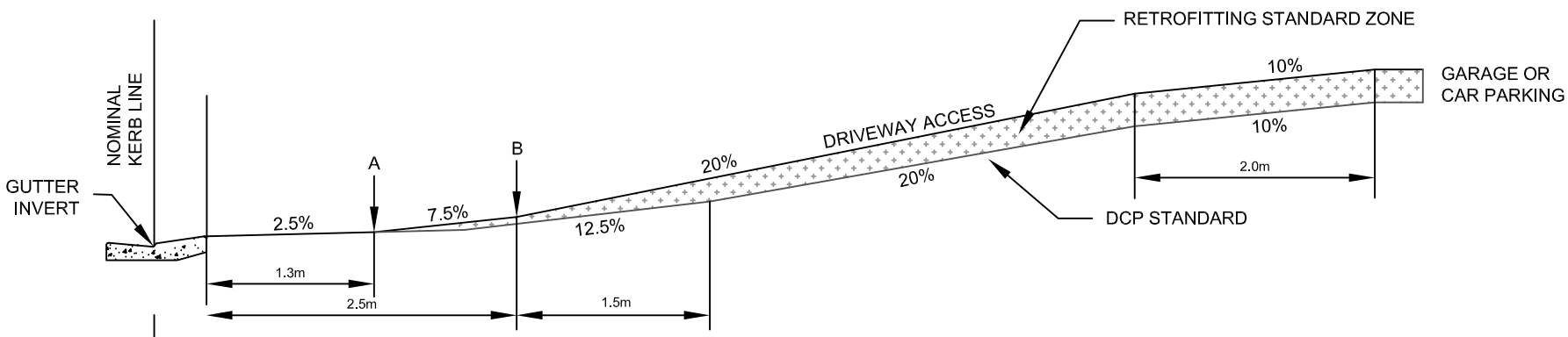
126-138 MAIN ROAD
SPEERS POINT
PHONE (02)4921 0333
www.lakemac.com.au

DRAWING TITLE
DRIVEWAY PROFILES - MAXIMUM GRADES WITHOUT FOOTPATH

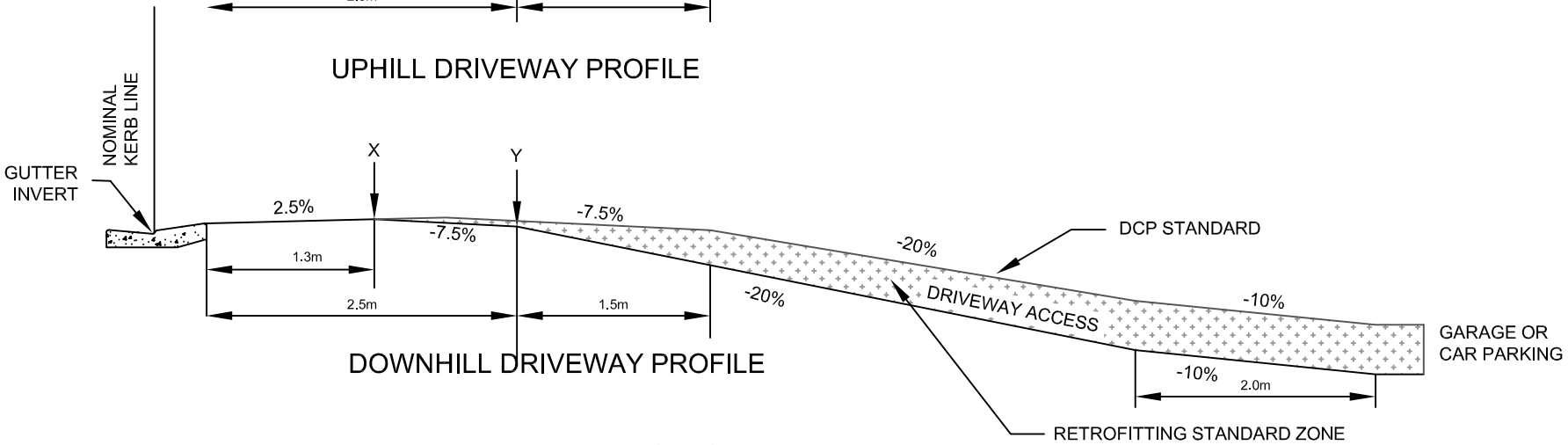
SHEET: 2 OF 3

DRAWING No. EGSD - 202

VERSION: 04



UPHILL DRIVEWAY PROFILE



DOWNHILL DRIVEWAY PROFILE

POINT A 1.3m BEHIND & 30mm ABOVE TOP OF LAYBACK

POINT B 2.5m BEHIND & 120mm ABOVE TOP OF LAYBACK

POINT X 1.3m BEHIND & 30mm ABOVE TOP OF LAYBACK

POINT Y 2.5m BEHIND & 60mm BELOW TOP OF LAYBACK

NOTES:

1. GRADING AND DIMENSIONS SHOWN SATISFY REQUIREMENTS OF AS/NZS 2890.1:2004 FOR B85 VEHICLES.
2. THE RECOMMENDED MAXIMUM CHANGE IN GRADE ACROSS A HINGE POINT SHOULD BE LIMITED TO 12.5% FOR A CONVEX TRANSITION OR 15% FOR A CONCAVE TRANSITION.
3. THE MAXIMUM GRADIENT OF A DOMESTIC DRIVE WAY WITHIN THE PROPERTY SHALL BE 1 in 5 (20%) IN ACCORDANCE WITH CURRENT DCP. WHERE EXISTING TOPOGRAPHY EXCEEDS 1 IN 5, THEN A MAXIMUM GRADE OF 1 IN 4 (25%) MAY BE USED WITH COUNCIL APPROVAL.
4. A 2.0m TRANSITION IS REQUIRED TO ANY PARKING AREA OR GARAGE AT A GRADE OF 10%.
5. WHERE AN ACCESS DRIVEWAY IS PROPOSED WITHIN AN OLDER DEVELOPED AREA AND WHERE A LARGE DIFFERENCE IN LEVEL BETWEEN THE EXISTING ROADWAY AND THE PROPERTY BOUNDARY OCCURS, THEN A SITE SPECIFIC DESIGN MAY BE REQUIRED WHICH MAY INCLUDE LOWERING THE EXISTING FOOTPATH. THESE ARE NON-STANDARD TREATMENTS AND WILL REQUIRE DESIGN BY A SUITABLY QUALIFIED PROFESSIONAL.
6. THE GRADES INDICATED WILL PROVIDE SATISFACTORY VEHICULAR ACCESS FOR MOST STANDARD VEHICLES. PROPERTY OWNERS SHOULD BE AWARE THAT SOME DIFFICULTIES WITH ACCESS MAY STILL OCCUR WITH MODIFIED VEHICLES, EXOTIC SPORTS CARS, AND TRAILERS INCLUDING CARAVANS AND BOAT TRAILERS.
7. ANY PROFILE EXCEEDING ANY OF THESE MAXIMUM SLOPES WILL REQUIRE COUNCIL APPROVAL.
8. ALL DESIGNS WITHIN THE RETROFITTING STANDARD ZONE ARE ACCEPTABLE.

VERSION	DATE	COMMENTS	PAPER SIZE
01	06/08/16	ORIGINAL ISSUE	A4

WORK TO FIGURED DIMENSIONS - DO NOT SCALE. DO NOT RELY ON THESE STANDARD DRAWINGS AS THE EQUIVALENT OF, OR SUBSTITUTE FOR, PROJECT-SPECIFIC DESIGN & ASSESSMENT BY A QUALIFIED PROFESSIONAL.

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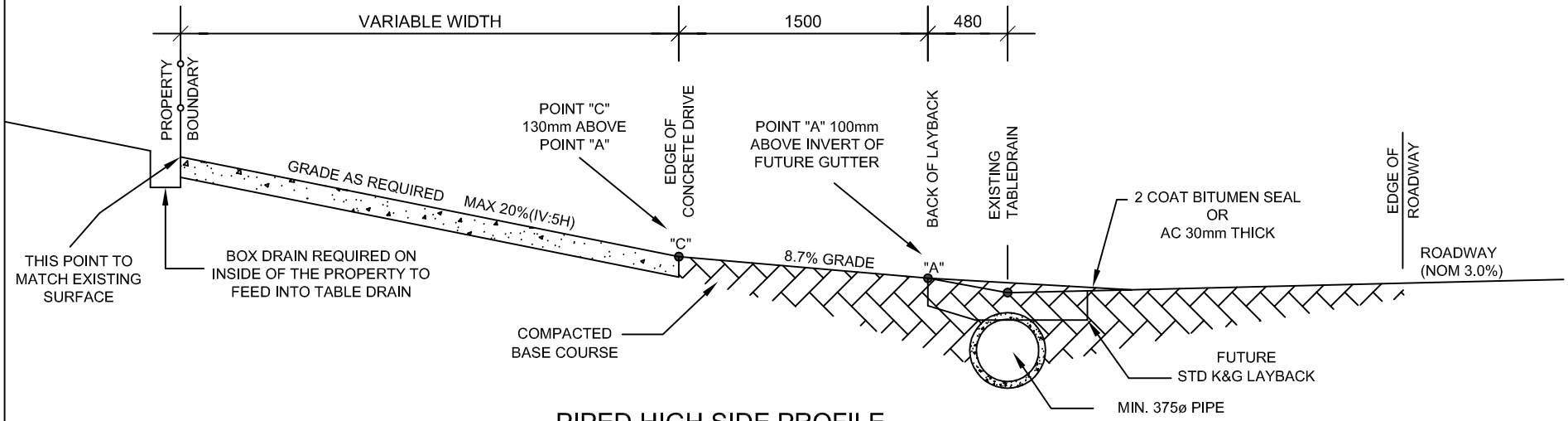
126-138 MAIN ROAD
SPEERS POINT
PHONE (02)4921 0333
www.lakemac.com.au

DRAWING TITLE: RETROFITTING DRIVEWAY - NON STANDARD SOLUTION WITHOUT FOOTPATH

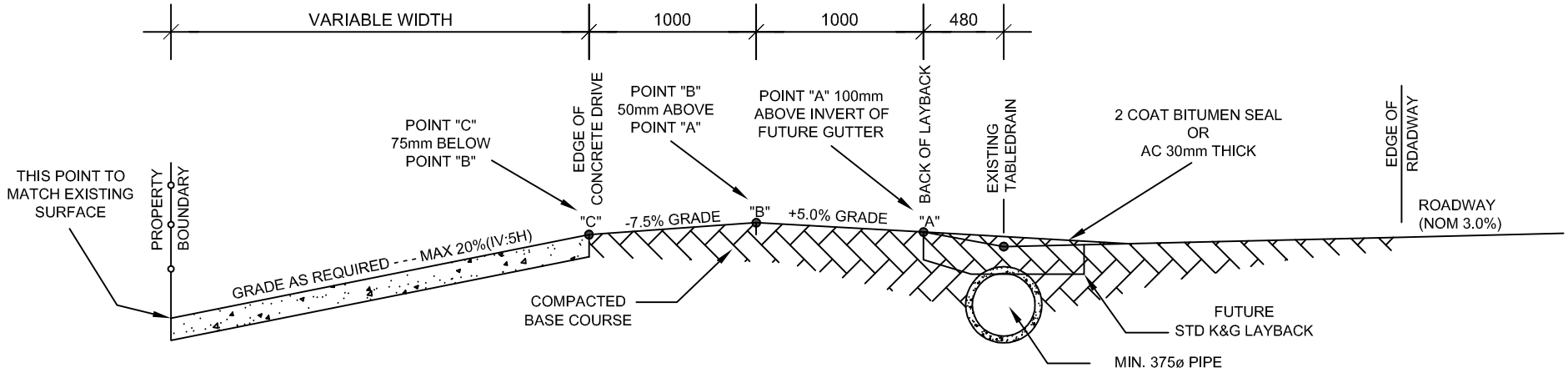
SHEET: 3 OF 3

DRAWING No. EGSD - 202

VERSION: 01



PIPED HIGH SIDE PROFILE



PIPED LOW SIDE PROFILE

NOTES:

1. WHILE COUNCIL DOES NOT NORMALLY PERMIT THE PLACEMENT OF CONCRETE OVER A PIPED ACCESS, IN CERTAIN CIRCUMSTANCES A "SACRIFICE" CONCRETE SLAB MAY BE PLACED BETWEEN "C" & EDGE OF ROADWAY. HOWEVER IF THE CONCRETE SLAB IS REMOVED DUE TO FUTURE ROADWORK, THERE WILL BE NO COMPENSATION PAID TO THE OWNER
2. TWO PRECAST HEADWALLS ARE REQUIRED TO BE INSTALLED ON EITHER SIDE OF CONCRETE
3. SMALLER SIZED PIPES ARE PERMITTED IN CERTAIN TERRAINS SUCH AS LOW LYING AREAS
4. ALL DIMENSIONS ARE IN MILLIMETRES

VERSION	DATE	COMMENTS	PAPER SIZE
04	29/07/16	AMENDMENT	A4
03	11/12/13	AMENDMENT	A4
02	29/05/13	AMENDMENT	A4
01	18/10/11	ORIGINAL ISSUE	A4

WORK TO FIGURED DIMENSIONS - DO NOT SCALE. DO NOT RELY ON THESE STANDARD DRAWINGS AS THE EQUIVALENT OF, OR SUBSTITUTE FOR, PROJECT SPECIFIC DESIGN AND ASSESSMENT BY A QUALIFIED PROFESSIONAL.

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126-138 MAIN ROAD
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DRAWING TITLE: RURAL DRIVEWAY MAXIMUM GRADES NO KERB & GUTTER

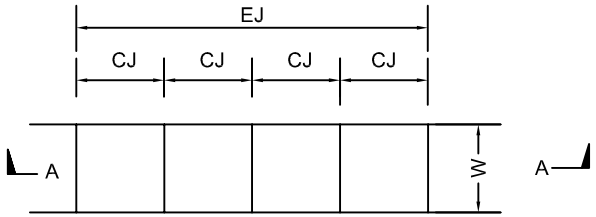
SHEET: 1 OF 1

DRAWING No. EGSD - 203

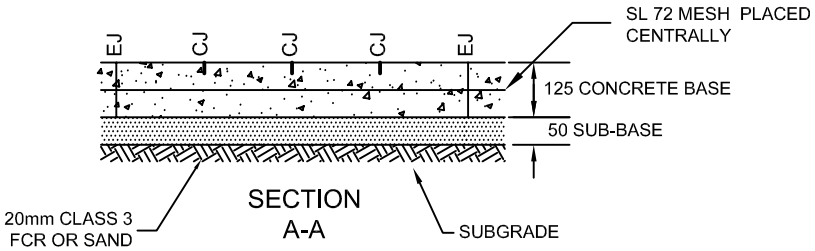
VERSION: 04

	SPACING
W	1200
CJ	1500
EJ	6000

SPACING OF CONTROL
& EXPANSION JOINTS



PLAN
FOOTPATH



SECTION
A-A

NOTES:

1. THE STANDARDS SHOWN ARE MINIMUM REQUIREMENTS ONLY
2. THE ALIGNMENT OF THE FOOTPATH IS TO BE IN ACCORDANCE WITH THE APPROVED ENGINEERING PLANS OR AS NOMINATED BY COUNCIL
3. FOR STANDARD CONCRETE FOOTPATH, CONCRETE TO BE REINFORCED WITH STEEL MESH TO ACHIEVE 25MPa WITH A MAXIMUM 20mm AGGREGATE SIZE
4. FOR COLOURED OR DECORATIVE FINISHES OR PATHS, 32MPa CONCRETE IS TO BE USED
5. FOR HAND PLACING OF CONCRETE - SLUMP TO BE 80mm
6. ALL FINISHED SURFACES ARE TO BE NON-SLIP AND MAINTAIN AN EVEN SURFACE COMPLYING WITH AS4586 AND AS3661.1
7. A MINIMUM OF 50mm COMPACTED 10mm SIZE CLASS 3 FCR OR SAND IS TO BE USED FOR THE SUB-BASE WHERE SOFT GROUND IS PRESENT A THICKER COMPACTED LAYER OF 75mm TO 100mm SHOULD BE USED
8. SUB-BASE TO BE LAID ON APPROVED COMPACTED NATURAL SUB-GRADE UNSUITABLE SUB-GRADE IS TO BE REMOVED AND REPLACED
9. MAXIMUM CROSSFALL ON ALL CONCRETE PATHS IS TO BE 2.5% (1 IN 40) IN ACCORDANCE WITH AS 1428 REQUIREMENTS
10. SPACING OF CONTROL & EXPANSION JOINTS TO BE AS PER TABLE. EXPANSION JOINTS TO BE CONNOLLY EXPANSION JOINTS (EXJ125) OR EQUIVALENT APPROVED BY COUNCIL
11. CONNOLLY JOINTS (OR APPROVED EQUIVALENT) ARE TO BE INSTALLED IN ACCORDANCE WITH THE MANUFACTURER'S INSTRUCTIONS OR GUIDELINES
12. ALL CONCRETE TO BE CURED BY USE OF PLASTIC SHEETING, DAMP SAND OR COMMERCIAL CURING COMPOUNDS
13. ALL JOINTS TO BE SMOOTH WITH MINIMAL IRREGULARITIES. CONCRETE MUST BE FINISHED FLUSH WITH THE TOP OF EACH JOINT SECTION
14. FINISHED PATH SURFACES, SHALL NOT DEVIATE BY MORE THAN 5MM ON 3M STRAIGHT EDGE
15. DISTURBED AREAS ADJACENT TO THE FOOTPATH ARE TO BE TURFED
16. WHERE PATH CROSSES VEHICULAR CROSSINGS, THICKNESS IS TO BE IN ACCORDANCE WITH STANDARD DRAWINGS FOR VEHICULAR CROSSING EGSD-104, OR EGSD-201
17. FOR KERB RAMPS REFER STANDARD DRAWING EGSD-102 REFERENCE MAY ALSO BE MADE TO LMCC ENGINEERING CONSTRUCTION GUIDELINES SECTION 0319 MINOR CONCRETE WORKS
18. IN NEW CONSTRUCTION INVOLVING A CONCRETE FOOTPATH 150x50x4 GALVANISED RHS TO BE INSTALLED UNDER THE CONCRETE FOOTPATH AS WELL AS THROUGH KERB
19. NON STANDARD TREATMENTS WILL BE SUBJECT TO APPROVAL BY COUNCIL
20. ALL DIMENSIONS ARE IN MILLIMETRES

VERSION	DATE	COMMENTS	PAPER SIZE
06	29/07/16	AMENDMENT	A4
05	21/11/13	AMENDMENT	A4
04	29/05/13	AMENDMENT	A4
03	03/05/13	AMENDMENT	A4
02	04/07/12	AMENDMENT	A4

WORK TO FIGURED DIMENSIONS - DO NOT SCALE. DO NOT RELY ON THESE STANDARD DRAWINGS AS THE EQUIVALENT OF, OR SUBSTITUTE FOR, PROJECT SPECIFIC DESIGN AND ASSESSMENT BY A QUALIFIED PROFESSIONAL.

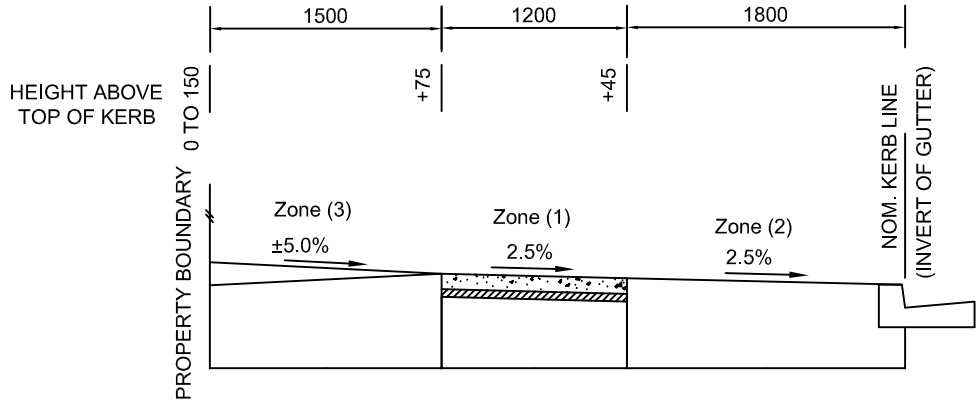
Lake Macquarie City Council
126-138 MAIN ROAD
SPEERS POINT
PHONE (02) 4921 0333
www.lakemac.com.au

DRAWING TITLE
CONCRETE FOOTPATHS
FOR NEW SUBDIVISIONS

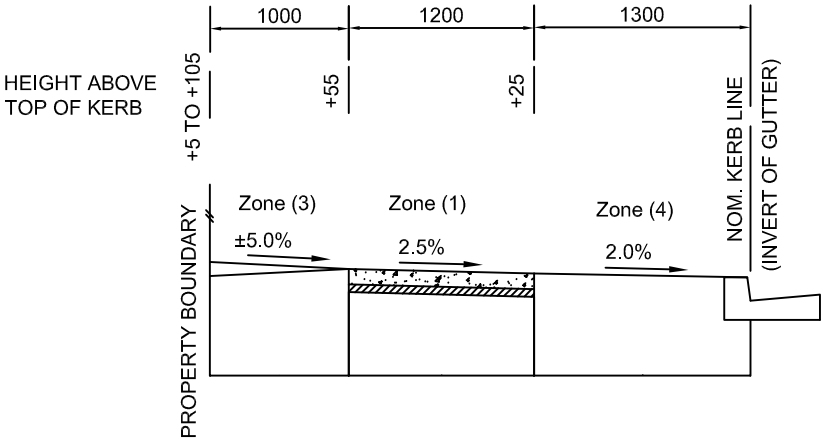
SHEET: 1 OF 3

DRAWING No. EGSD - 301

VERSION: 06



LOCATION OF CONCRETE FOOTPATH
WITHIN A 4.5m FOOTWAY



LOCATION OF CONCRETE FOOTPATH
WHEN REQUIRED TO BE RETRO-FITTED
WITHIN A 3.5m FOOTWAY

NOTES:

1. WHEREVER POSSIBLE THE GRADIENTS SHOWN FOR ZONES (1), (2), (3) AND (4) SHOULD BE USED
   - ZONE (1) THE MAXIMUM CROSSFALL ACROSS THE CONCRETE FOOTPATH IS 2.5%. (1 IN 40)
   - ZONE (2) THIS AREA IS ALSO BEST GRADED AT 2.5% (1 IN 40) BUT A MAXIMUM OF 5% (1 IN 20) CAN BE USED
   - ZONE (3) A MAXIMUM OF ±5% (1 IN 20) IS RECOMMENDED. THIS IS A SERVICE CORRIDOR AND SEVERE CHANGES IN GRADE ARE NOT ACCEPTABLE
   - ZONE (4) THE REDUCTION TO 2.0% (1 IN 50) IS NECESSARY TO INCORPORATE A KERB RAMP AT 1 IN 8 MAX. GRADIENT
2. ALL DIMENSIONS ARE IN MILLIMETRES

VERSION	DATE	COMMENTS	PAPER SIZE
06	29/07/16	AMENDMENT	A4
05	11/12/13	AMENDMENT	A4
04	29/05/13	AMENDMENT	A4
03	03/05/13	AMENDMENT	A4
02	04/07/12	AMENDMENT	A4

WORK TO FIGURED DIMENSIONS - DO NOT SCALE. DO NOT RELY ON THESE STANDARD DRAWINGS AS THE EQUIVALENT OF, OR SUBSTITUTE FOR, PROJECT SPECIFIC DESIGN AND ASSESSMENT BY A QUALIFIED PROFESSIONAL.

Lake Macquarie City Council
126-138 MAIN ROAD
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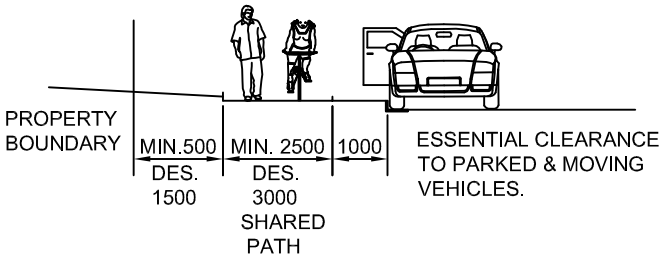
DRAWING TITLE: CONCRETE FOOTPATHS CROSS-SECTION

SHEET: 2 OF 3

DRAWING No. EGSD - 301

VERSION: 06

* DES = DESIRABLE



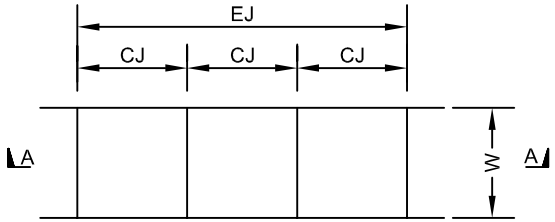
DESIRABLE CLEARANCE OF SHARED PATH TO ROAD

	SPACING
W	2500 / 3000
CJ	4000
EJ	12000

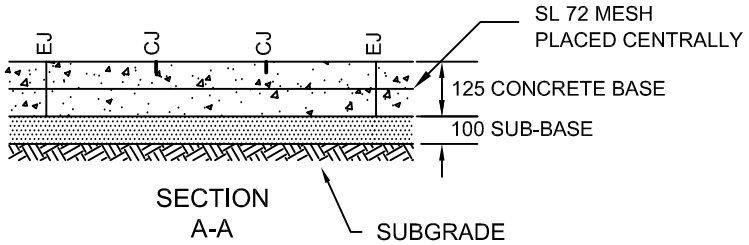
SPACING OF CONTROL & EXPANSION JOINTS

W = 2500
MINIMUM WIDTH
OF SHARED PATH

W = 3000
DESIRABLE WIDTH
FOR SHARED
PATH/CYCLEWAY



PLAN
SHARED PATH



SECTION
A-A

NOTES:

1. SHARED PATHS SIGNAGE SHALL BE PROVIDED AND LOCATED GENERALLY IN ACCORDANCE WITH AUSTROADS AND RMS STANDARDS
2. SHARED PATH HANDRAILS TO BE PROVIDED AT ROAD CROSSINGS WHERE SHOWN ON THE APPROVED ENGINEERING PLANS OR WHERE NOMINATED BY COUNCIL
3. CONCRETE FOR SHARED PATHS TO BE REINFORCED WITH STEEL MESH TO ACHIEVE STRENGTH 25MPa WITH A MAXIMUM 20mm AGGREGATE SIZE
4. CONSIDERATION WILL BE GIVEN TO FIBRE REINFORCED CONCRETE AND WHERE APPROVED, IS TO ACHIEVE FLEXURAL STRENGTH 5.5MPa AT 28 DAYS
5. FOR HAND PLACING OF CONCRETE - SLUMP TO BE 80mm
6. ALL FINISHED SURFACES ARE TO BE NON-SLIP AND MAINTAIN AN EVEN SURFACE COMPLYING WITH AS4586 AND AS3661.1
7. A MINIMUM OF 100mm COMPACTED 20mm CLASS 3 FCR TO BE USED FOR THE SUB-BASE. WHERE SOFT GROUND IS PRESENT A THICKER COMPACTED LAYER SHOULD BE USED
8. SUB-BASE TO BE LAID ON APPROVED COMPACTED NATURAL SUB-GRADE UNSUITABLE SUB-GRADE IS TO BE REMOVED AND REPLACED
9. MAXIMUM CROSSFALL ON ALL CONCRETE PATHS IS TO BE 2.5% (1 IN 40) IN ACCORDANCE WITH AS 1428 REQUIREMENTS
10. SPACING OF CONTROL & EXPANSION JOINTS TO BE AS PER TABLE. EXPANSION JOINTS TO BE CONNOLLY EXPANSION JOINTS (EXJ125) OR EQUIVALENT APPROVED BY COUNCIL
11. CONNOLLY JOINTS (OR APPROVED EQUIVALENT) ARE TO BE INSTALLED IN ACCORDANCE WITH THE MANUFACTURER'S INSTRUCTIONS OR GUIDELINES
12. ALL CONCRETE TO BE CURED BY USE OF PLASTIC SHEETING, DAMP SAND OR COMMERCIAL CURING COMPOUNDS
13. ALL JOINTS TO BE SMOOTH WITH MINIMAL IRREGULARITIES. CONCRETE MUST BE FINISHED FLUSH WITH THE TOP OF EACH JOINT SECTION
14. FINISHED PATH SURFACES, SHALL NOT DEVIATE BY MORE THAN 5MM ON 3M STRAIGHT EDGE
15. DISTURBED AREAS ADJACENT TO THE FOOTPATH ARE TO BE TURFED
16. WHERE PATH CROSSES VEHICULAR CROSSINGS, THICKNESS IS TO BE IN ACCORDANCE WITH STANDARD DRAWINGS FOR VEHICULAR CROSSING EGSD-104, OR EGSD-20
17. FOR KERB RAMPS REFER STANDARD DRAWING EGSD-102 REFERENCE MAY ALSO BE MADE TO LMCC ENGINEERING CONSTRUCTION GUIDELINES SECTION 0319 MINOR CONCRETE WORKS
18. IN NEW CONSTRUCTION INVOLVING A CONCRETE SHARED PATH GALVANISED RHS 150x50x4 TO BE INSTALLED UNDER THE CONCRETE FOOTPATH AS WELL AS THROUGH KERB
19. NON STANDARD TREATMENTS WILL BE SUBJECT TO APPROVAL BY COUNCIL
20. ALL DIMENSIONS ARE IN MILLIMETRES

VERSION	DATE	COMMENTS	PAPER SIZE
06	29/07/16	AMENDMENT	A4
05	21/11/13	AMENDMENT	A4
04	29/05/13	AMENDMENT	A4
03	03/05/13	AMENDMENT	A4
02	04/07/12	AMENDMENT	A4

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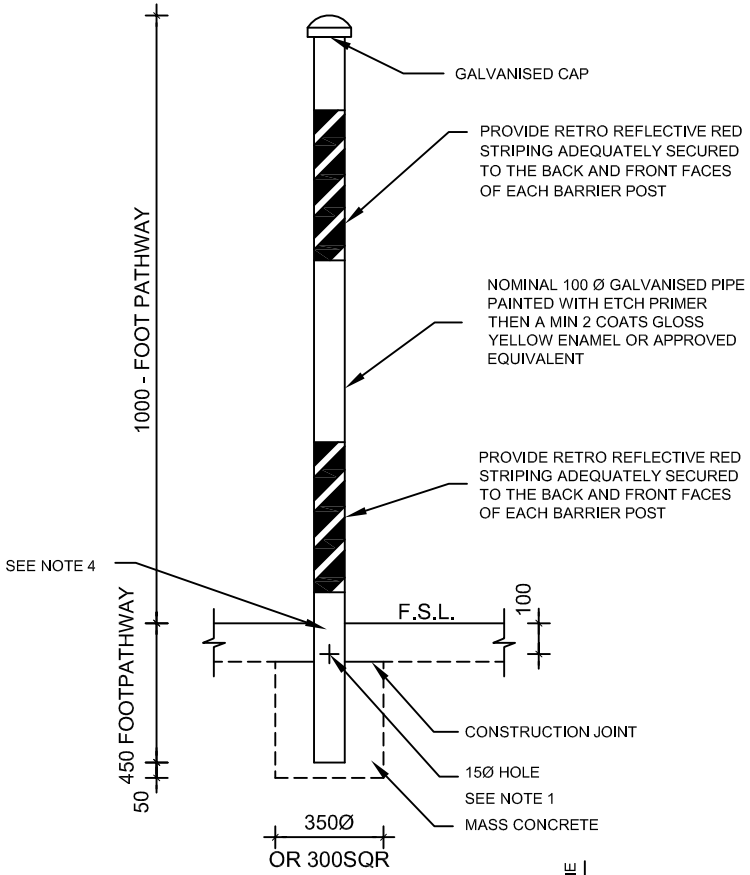
Lake Macquarie City Council
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DRAWING TITLE
CONCRETE SHARED PATH

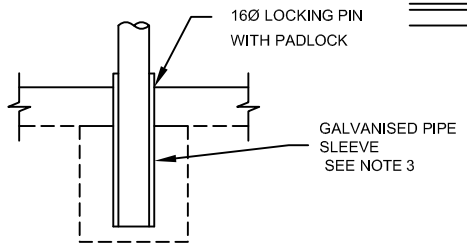
SHEET: 3 OF 3

DRAWING No. EGSD - 301

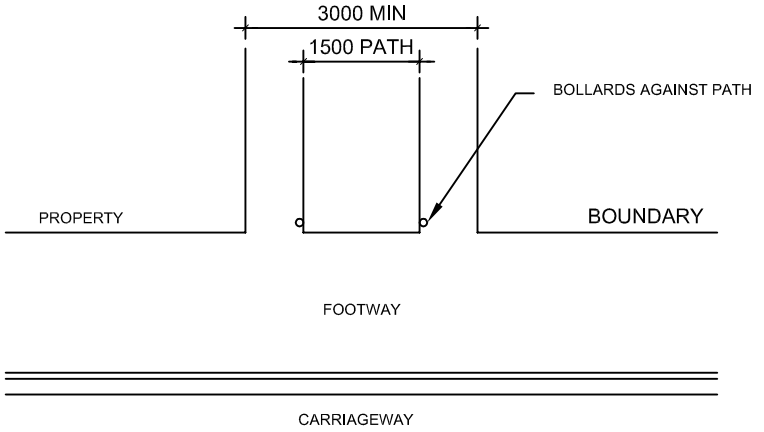
VERSION: 06



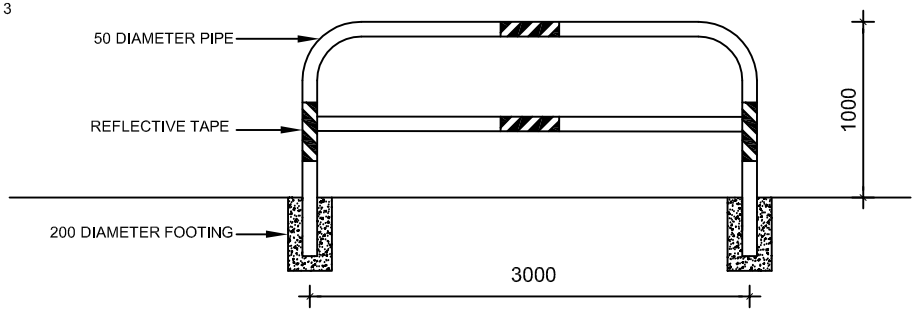
BOLLARD DETAIL

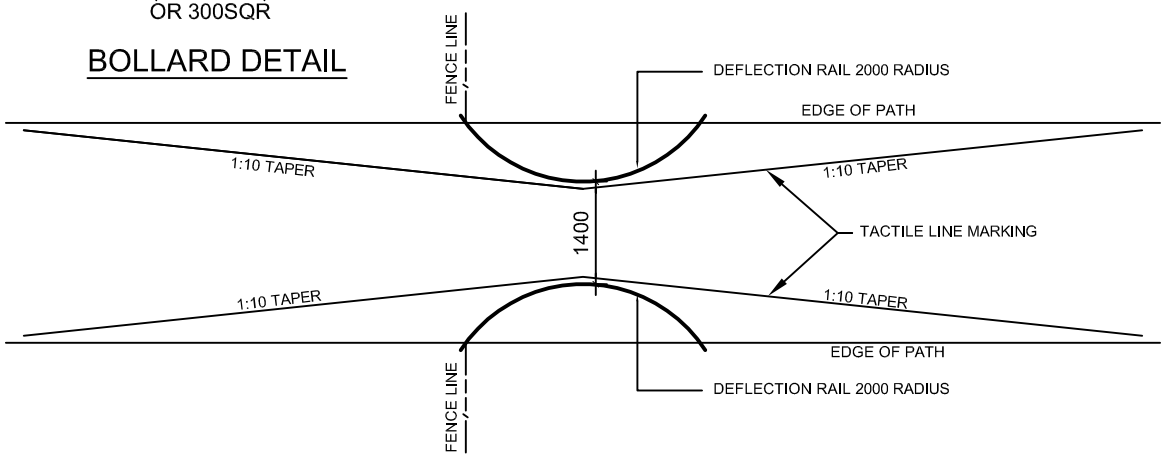


SLEEVE DETAIL (FOR REMOVABLE BOLLARD ONLY)



FOOT PATH PLAN





SHARED PATH PLAN DEFLECTION RAIL

NOTES:

1. PLACE 150mm N12 BAR IN HOLE BEFORE FIXING INTO PLACE.
2. ALL STEEL TO BE HOT DIPPED GALVANISED OR APPROVED EQUIVALENT.
3. WHERE PATHWAYS ARE USED FOR MAINTENANCE OR EMERGENCY VEHICLE ACCESS, BOLLARDS AT ONE END OF THE PATH ARE TO BE REMOVABLE WITH LOCKING PIN.
4. CONSIDERATION TO BE GIVEN TO 1800mm HIGH BOLLARDS AND 60mm DEEP FOOTINGS FOR SHARED PATHWAYS.
5. ALL DIMENSIONS IN MILLIMETRES.

VERSION	DATE	COMMENTS	PAPER SIZE
04	29/07/16	AMENDMENT	A4
03	25/06/15	AMENDMENT	A4
02	29/05/13	AMENDMENT	A4
01	19/10/11	ORIGINAL ISSUE	A4

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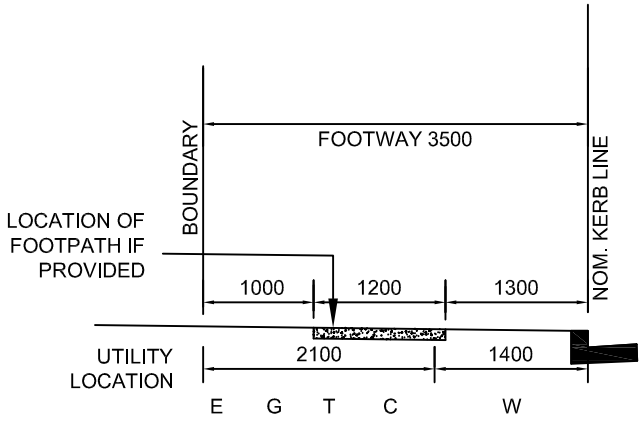
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DRAWING TITLE: PATHWAY BARRIERS - BOLLARDS

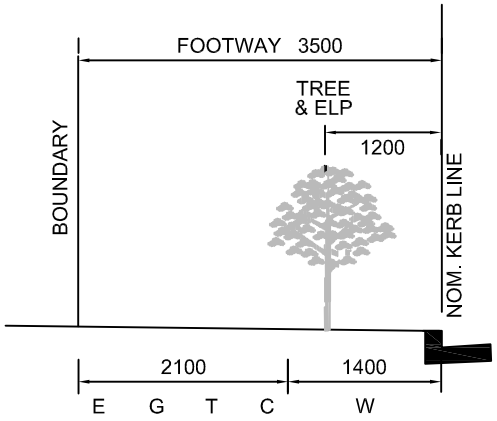
SHEET: 1 OF 1

DRAWING No. EGSD-302

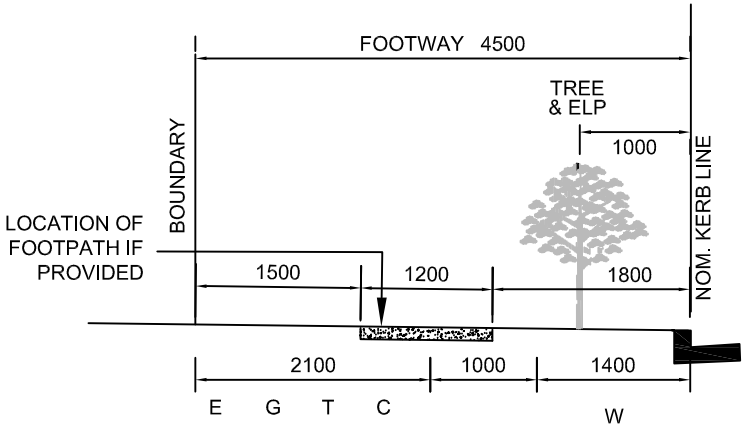
VERSION: 04



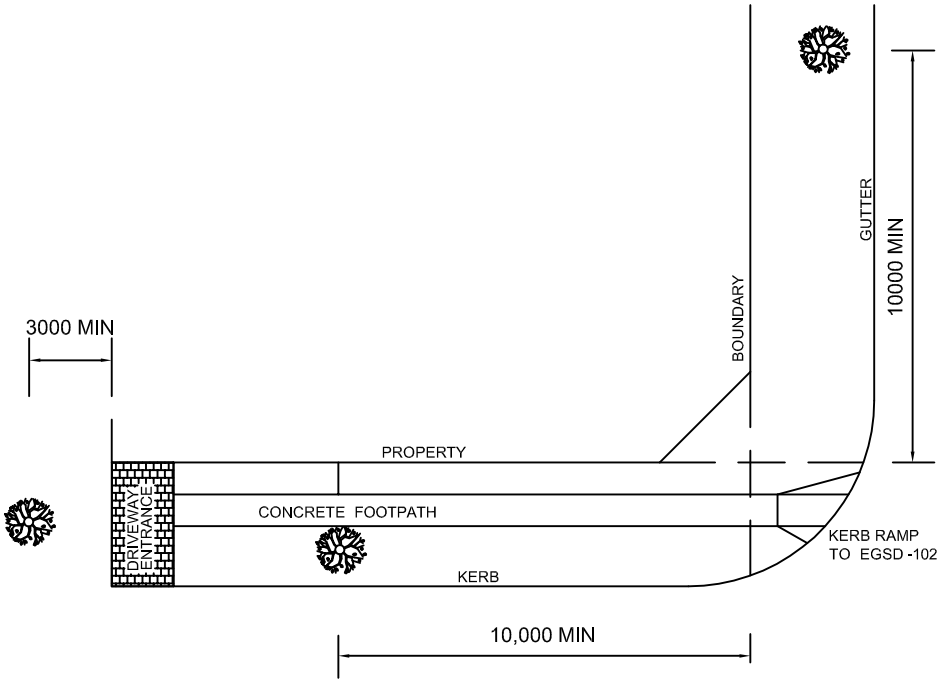
FOOTWAY 3.5m WIDE



FOOTWAY 3.5m WIDE WITHOUT FOOTPATH



FOOTWAY 4.5m WIDE



CLEARANCE REQUIREMENTS PLAN

LEGEND

E - ELECTRICITY
G - GAS
T - TELEPHONE
C - COMMUNICATIONS - MAIN CABLE
W - WATER

NOTES:

1. TREES PLANTED WITHIN THE FOOTWAY MUST HAVE ROOT BARRIERS INSTALLED AT THE TIME OF PLANTING
2. TREES LIMITED TO A MAXIMUM HEIGHT OF 3m ARE TO BE GROWN UNDER POWER LINES
3. CONSULTATION WITH PUBLIC UTILITY AUTHORITIES VIA DIAL BEFORE YOU DIG FOR LOCATION OF SERVICES SHOULD OCCUR PRIOR TO ANY EXCAVATION
4. IN OLDER AREAS PUBLIC UTILITIES MAY OCCUPY DIFFERENT ALIGNMENTS TO THOSE SHOWN ON THIS DRAWING
5. REFER TO LMCC LANDSCAPE GUIDELINES FOR TREE PLANTING DETAILS
6. FOOTWAY PLANTING IS GENERALLY ONLY APPROPRIATE WHERE THE FOOTWAY WIDTH EXCEEDS 3.5m OR IF THERE IS NO FOOTPATH
7. PLANTING IN A FOOTWAY LESS THAN 3.5m WIDE REQUIRES COUNCIL APPROVAL
8. RETRO FITTING WITHIN AN EXISTING 3.5m WIDE FOOTWAY REQUIRES COUNCIL APPROVAL BUT IS TO BE CONSISTENT WITH THE NSW STREETS OPENING CONFERENCE "GUIDE TO CODES AND PRACTICES FOR STREET OPENINGS" FOOTWAY ALLOCATION FOR LMCC
9. ALL DIMENSIONS ARE IN MILLIMETRES

VERSION	DATE	COMMENTS	PAPER SIZE
04	05/08/16	AMENDMENT	A4
03	11/12/13	AMENDMENT	A4
02	29/05/13	AMENDMENT	A4
01		ORIGINAL ISSUE	A4

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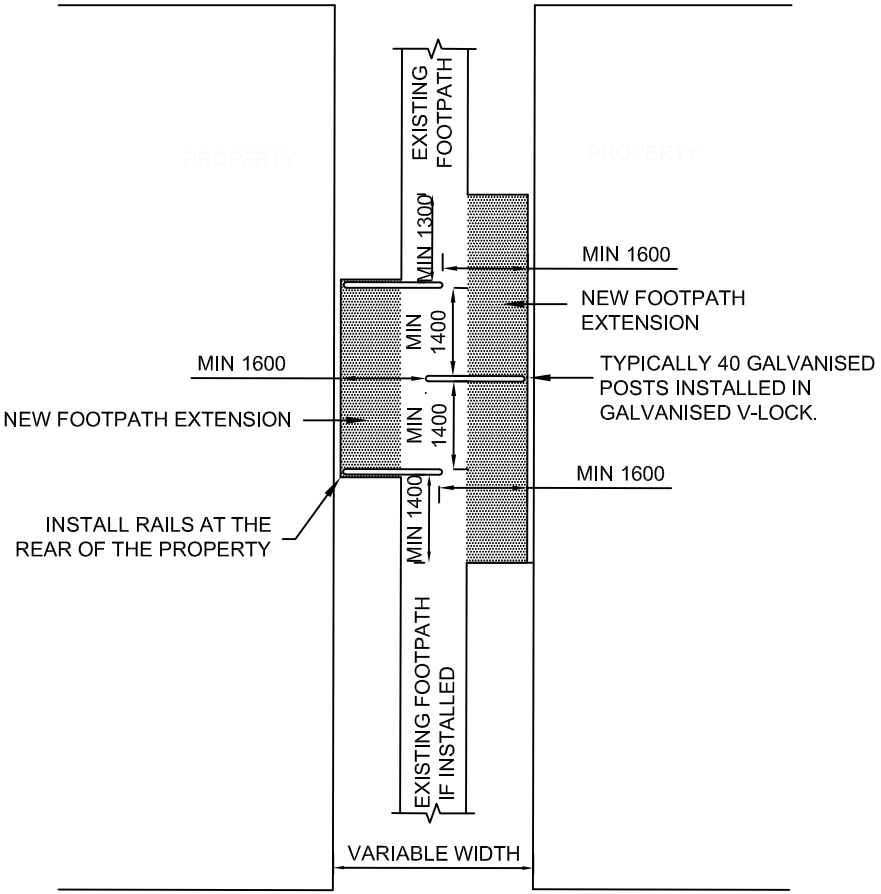
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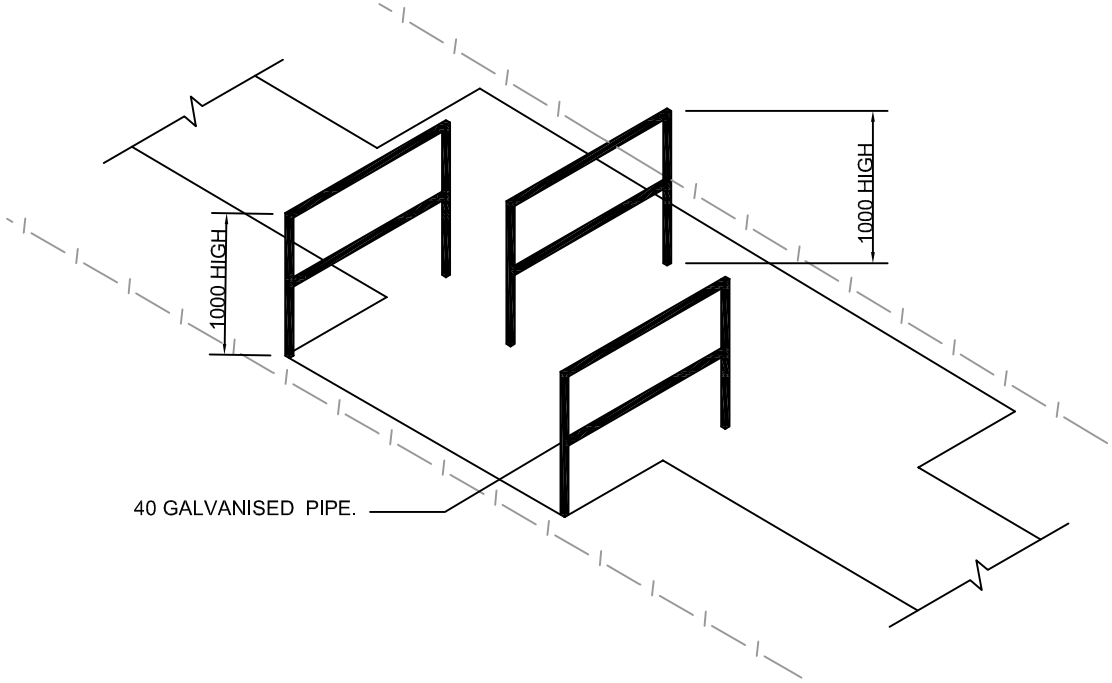
DRAWING TITLE
FOOTWAY ALLOCATION
UTILITY SERVICES AND TREES

SHEET: 1 OF 1

DRAWING No. EGSD-303

VERSION: 04





NOTES:

1. VERIFY PUBLIC UTILITY LOCATIONS BEFORE ANY EXCAVATION
2. IMPLEMENT EROSION AND SEDIMENT CONTROL MEASURES IN ACCORDANCE WITH THE LMCC ENGINEERING GUIDELINES
3. READ IN CONJUNCTION WITH THE LMCC ENGINEERING GUIDELINES
4. WIDEN SECTIONS OF FOOTPATH, MAXIMUM SLOPE TO EXISTING FOOTPATH 1 IN 40
5. PEDESTRIAN RAILS WILL BE VARIABLE WIDTH DEPENDING ON SITE CONDITIONS, ALL RAILS ARE TO BE 1000mm HIGH, SET INTO V-LOCKS
6. REFLECTOR TAPE TO BE INSTALLED AT ALL CORNERS AND IN THE MIDDLE OF THE HORIZONTAL RAILS
7. ALL DIMENSIONS ARE IN MILLIMETRES

VERSION	DATE	COMMENTS	PAPER SIZE
03	29/07/16	UPDATE NOTES	A4
02	25/06/15	UPDATE NOTES	A4
01	27/02/14	ORIGINAL ISSUE	A4

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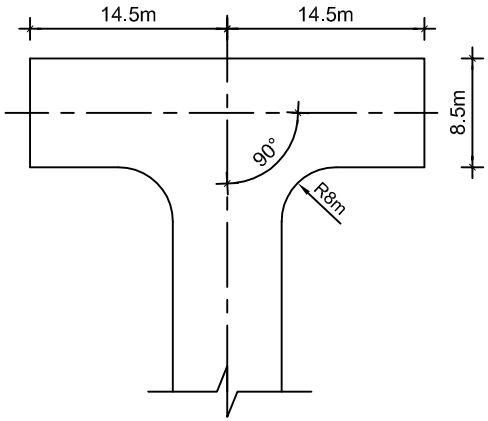
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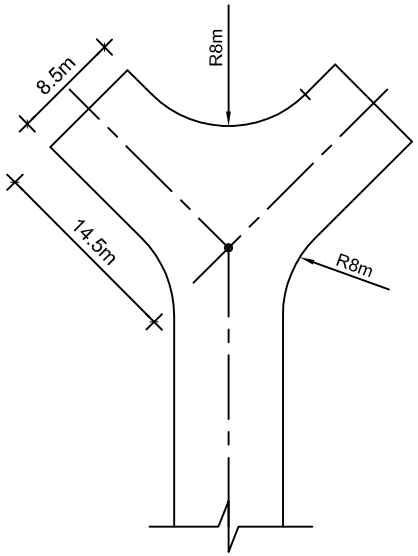
DRAWING TITLE: FOOTPATH CHICANE

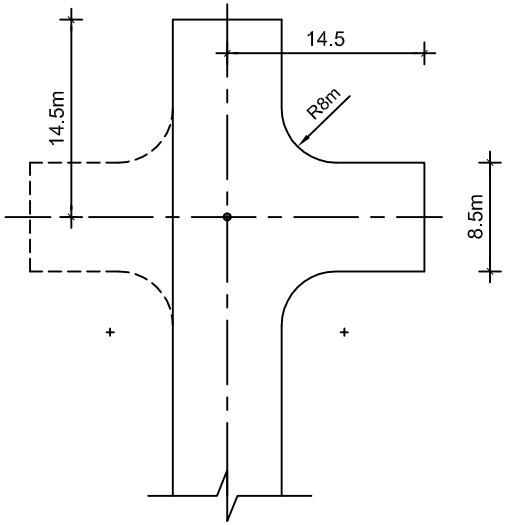
SHEET: 1 OF 1

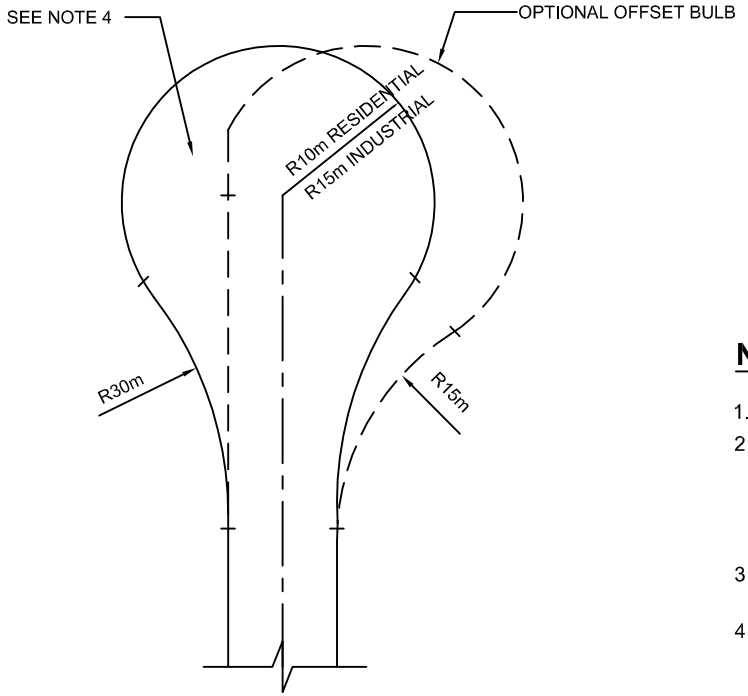
DRAWING No.: EGSD-304

VERSION: 03









NOTES:

1. INCOMING WIDTHS TO SUIT ROAD CATEGORIES
2. NOTWITHSTANDING THE ABOVE DIMENSIONING, AUSTRALIAN STANDARD HB 72-1195 FOR HEAVY RIGID VEHICLES (HRV) SHALL BE USED TO VERIFY ADEQUATE MANOEUVERING AREA
3. VARIATIONS TO THESE MAY BE CONSIDERED DEPENDING ON CIRCUMSTANCES
4. TEMPORARY BULB TO BE MIN. 7m R. ONE COAT SEAL ON APPROVED PAVEMENT

VERSION	DATE	COMMENTS	PAPER SIZE
02	06/08/11	AMENDMENT	A4
01	19/10/11	ORIGINAL ISSUE	A4

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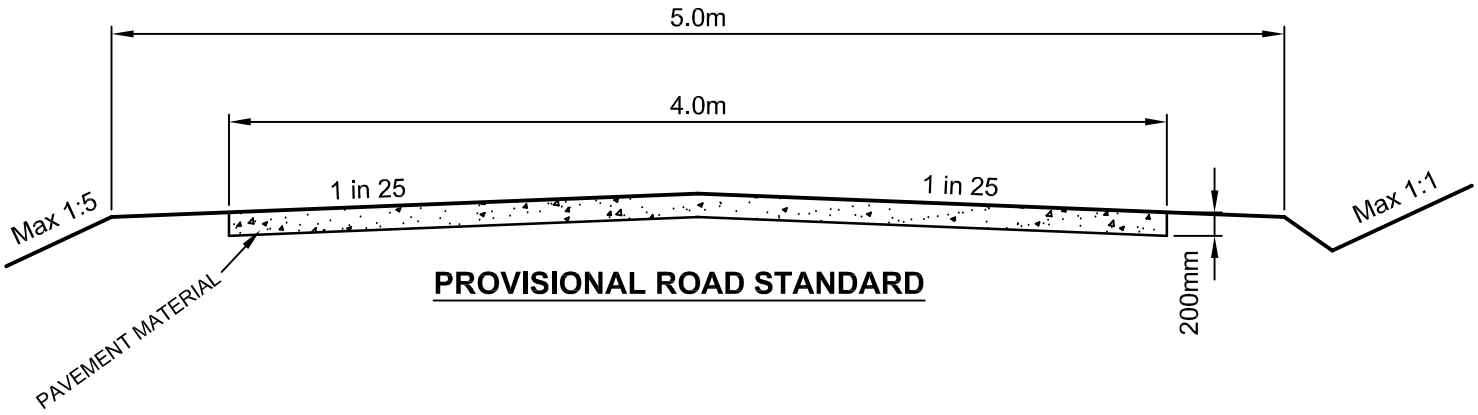
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DRAWING TITLE: TURNING AREAS FOR CUL-DE-SACS

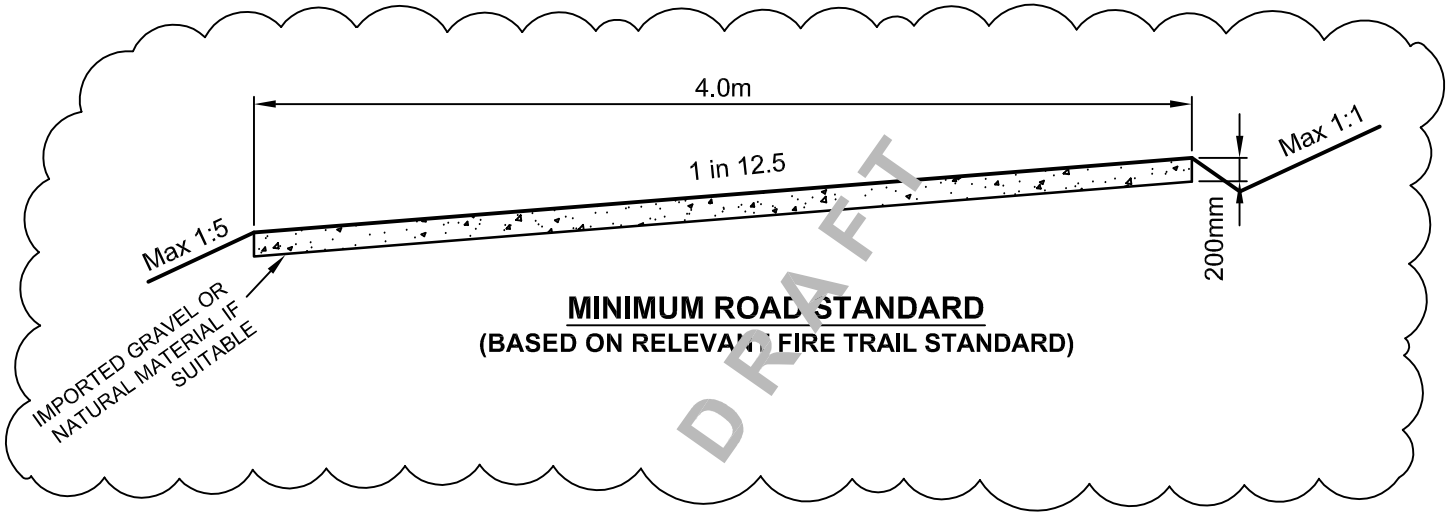
SHEET: 1 OF 1

DRAWING No. EGSD-701

VERSION: 02



PROVISIONAL ROAD STANDARD



MINIMUM ROAD STANDARD
(BASED ON RELEVANT FIRE TRAIL STANDARD)

DRAFT

NOTES:

PROVISIONAL ROAD STANDARD

1. PAVEMENT MATERIAL IS TO BE 200mm NOMINAL CLASS 2 CRUSHED ROCK AND BE 200MM CONSOLIDATED DEPTH
2. IF ANY ROAD CENTRE LINE GRADIENT IS GREATER THAN 1:8, THEN THAT SECTION OF ROAD IS TO BE SEALED WITH A TWO COAT BITUMEN SEAL, AFTER SATISFACTORY BENKLEMANN BEAM TESTING OF THE BASE COURSE
3. TABLE DRAINS ARE TO BE PROVIDED ON EACH SIDE OF THE ROAD.
4. PIPED CULVERTS ARE TO BE INSTALLED WHERE NATURAL DEPRESSIONS OR WATERCOURSES CROSS THE ROAD. THE PIPE MUST BE DESIGNED TO COPE WITH A 1:5 YEAR FLOOD FREQUENCY. MINIMUM PIPE SIZE TO BE USED IS 375MM
5. IF THE ACCESS ROAD CROSSES A FLOODWAY AREA THEN THE DESIGN MUST CATER FOR A 1:20 YEAR FLOOD FREQUENCY. DEPTH MARKING SIGNAGE ARE REQUIRED

MINIMUM ROAD STANDARD

6. PASSING BAYS ARE TO BE LOCATED AT 500m INTERVALS WHERE POSSIBLE.
7. GRADES ARE TO BE LESS THAN 25%.
8. SPACING OF CONTOUR BANKS IS TO BE IN ACCORDANCE WITH BUSH FIRE COORDINATING COMMITTEE POLICY No. 2/2007.
9. DRAINAGE TO SUIT LOCAL TERRAIN.
10. ALL FIRE TRAIL MANAGEMENT IS TO CONFORM WITH BUSH FIRE COORDINATING COMMITTEE POLICY No. 2/2007.

VERSION	DATE	COMMENTS
02	06/08/16	AMENDMENT
01	17/06/13	ORIGINAL ISSUE

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DRAWING TITLE: PROVISIONAL AND MINIMUM ROAD STANDARDS

SHEET: 1 OF 1

DRAWING No. EGSD-702

VERSION: 02

FOOTWAY RESTORATION CONDITIONS

RESTORATION CONDITIONS FOR NATURAL SURFACE, DRIVEWAY AND FORMED FOOTPATH

1. Hours of operation will be 7am to 5pm, Monday to Friday, unless otherwise stated.
2. The Permit Holder will advise Council of any change to location, scope of work, project dates, contact information, or any other details of the works as per the Permit Application.
3. All precautions will be taken to protect the public from personal injury and damage to property, including the erection of necessary barriers, lights and signs in accordance with Australian Standard 1742.3 while the work is in progress.
4. The Permit holder will be liable for any injury to persons or property resulting from a failure to comply with the conditions of the Permit.
5. Traffic control will be provided and maintained by the Permit Holder in accordance with RMS "Traffic Control at Work Sites" 2010 and AS 1742. Traffic Control Plans will be submitted to Council prior to commencement of works. Where approval is given to trench the road, only half the road is to be opened at a time.
6. Dial Before U Dig information (telephone 1100 or at www.dialbeforeyoudig.com.au) will be obtained for the details of Utility Authorities infrastructure at or near the proposed works. For the location of storm water pipes, contact Council.
7. Environmental controls will be put in place to prevent erosion and to stop silt or construction materials from leaving the site. Control measures will be in accordance with Australian Standards and environmental requirements.
8. Utility services will be located within the allocation as shown on Council Drawing EGSD-303 - "Footway allocation utility services and trees". Contact Council if this is not possible.
9. Natural surface areas will be reinstated with "like for like" materials, including topsoil and grass.

FOOTPATH/CYCLEWAY/ DRIVEWAY RESTORATIONS

1. The Permit Holder will backfill trenches using excavated materials, placing and thoroughly compacting the materials in 150mm layers.
2. The Permit Holder will maintain the integrity and appearance of the footpath, cycleway or driveway by providing "like for like" reinstatement. All joints will be straight cut with a concrete saw. Entire slabs will be removed - with the exception of driveways where, Council may approve a minimum of 1 metre wide strip (Council must be contacted)
3. Kerb ramps will be reinstated to Council standards - see Council's Standard Drawing EGSD-102.
4. Like for like" replacement - Standard existing footpaths are 1.2 metres wide and 80mm thick. Control joints are provided every 1.5 metres and expansion joints every 6.0 metres. The concrete is broom finished.
5. Cycleways or new footpaths should be restored/installed according to Council's Standard Drawing EGSD-301.
6. Standard driveways are 125mm thick with F72 Steel, dowelled into the existing concrete every 600mm EGSD-201.

TREES (Greater than 3m in height or 200MM in diameter)

1. Pruning or removal of trees requires written permission from Council.
2. All trees are to be retained in an undamaged condition unless written permission from Council for pruning or removal has been provided.
3. All trees to be protected in accordance with AS4970-2009, "Protection of Trees".
4. Where works occur within the drip line, or tree protection zone of a tree, a tree protection plan prepared by a suitably qualified arborist in accordance with AS4970-2009 is required.
5. If a tree goes into decline due to the works, the contractor will be responsible for any remediation or removal (at Council's discretion) of the tree for twelve months from the time the job has been completed.
6. Plans and survey drawings must show trees present. Such trees should be numbered sequentially on the plans and preferably include tags on the trees to avoid any confusion.
7. Where works are within proximity to trees, tree protection zones and structural root zones must be calculated and shown on the work plans.

VERSION	DATE	COMMENTS
01	06/08/2016	ORIGINAL ISSUE

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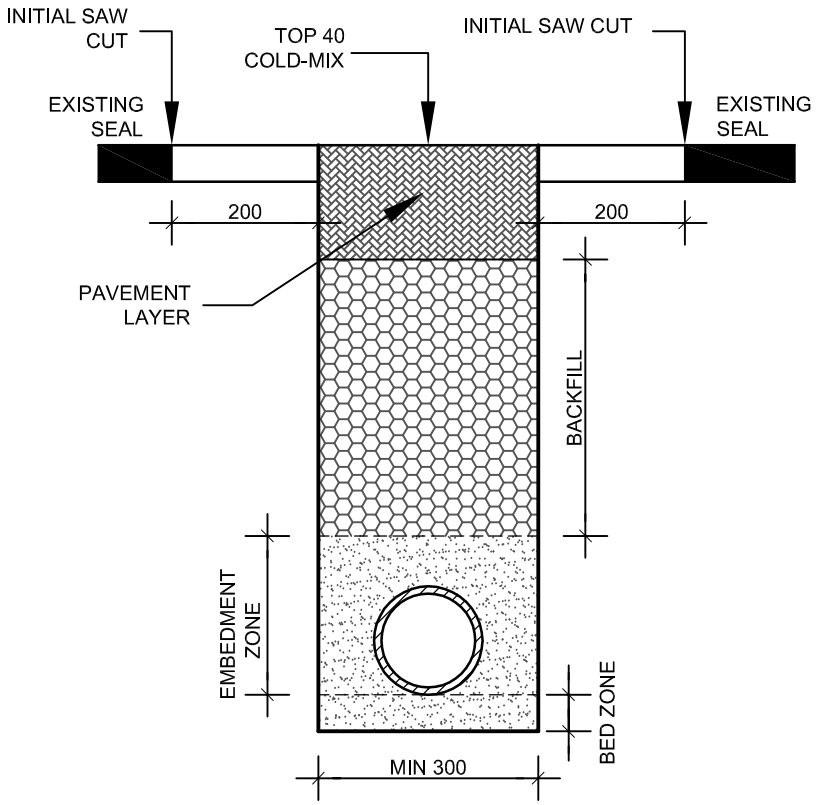
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DRAWING TITLE
SERVICE TRENCH
FOOTWAY RESTORATION

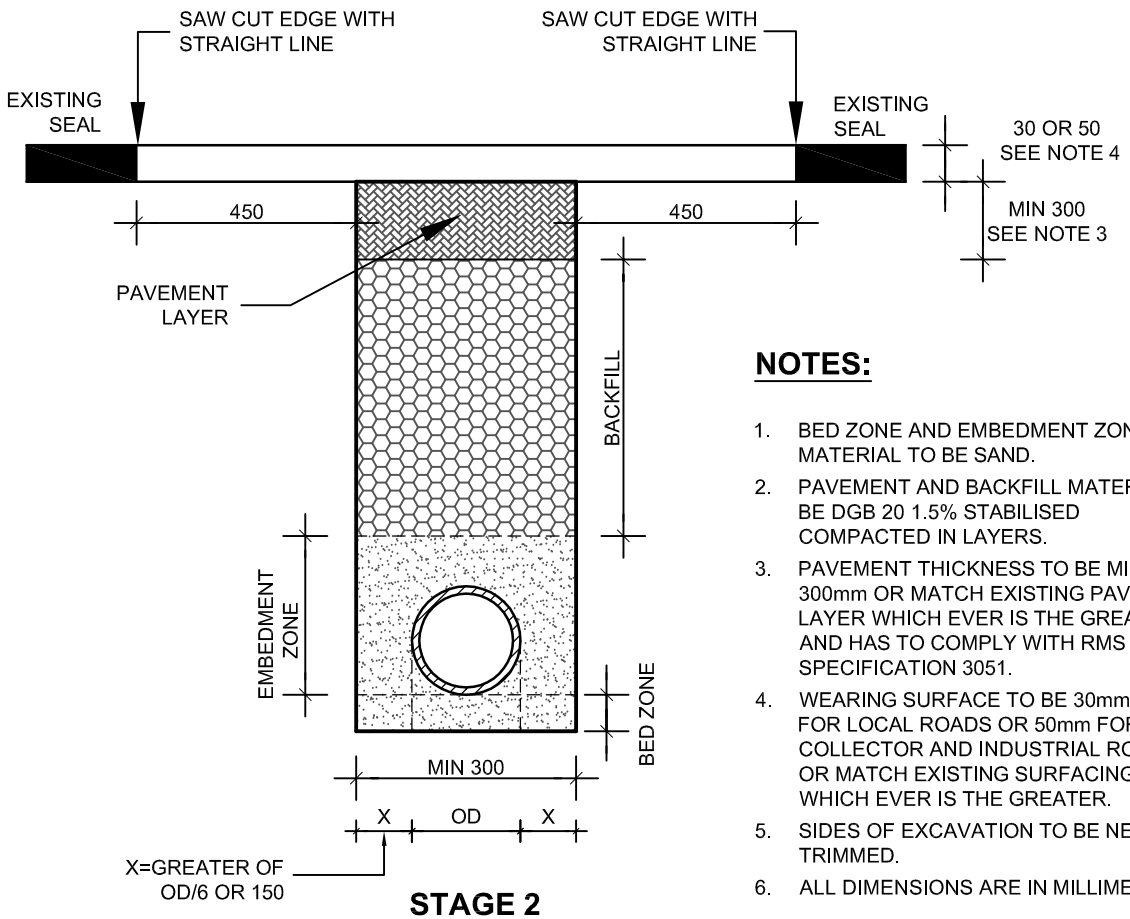
SHEET: 1 OF 1

DRAWING No. EGSD-703

VERSION: 01



STAGE 1



STAGE 2

NOTES:

1. BED ZONE AND EMBEDMENT ZONE MATERIAL TO BE SAND.
2. PAVEMENT AND BACKFILL MATERIAL TO BE DGB 20 1.5% STABILISED COMPACTED IN LAYERS.
3. PAVEMENT THICKNESS TO BE MIN 300mm OR MATCH EXISTING PAVEMENT LAYER WHICH EVER IS THE GREATER AND HAS TO COMPLY WITH RMS SPECIFICATION 3051.
4. WEARING SURFACE TO BE 30mm AC10 FOR LOCAL ROADS OR 50mm FOR COLLECTOR AND INDUSTRIAL ROADS OR MATCH EXISTING SURFACING LAYER WHICH EVER IS THE GREATER.
5. SIDES OF EXCAVATION TO BE NEATLY TRIMMED.
6. ALL DIMENSIONS ARE IN MILLIMETRES.

VERSION	DATE	COMMENTS
01	06/08/2016	ORIGINAL ISSUE

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126-138 MAIN ROAD
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DRAWING TITLE: SERVICE TRENCH ROAD RESTORATION UP TO 1.2m WIDTH

SHEET: 1 OF 2

DRAWING No. EGSD-704

VERSION: 01

ROADWAY RESTORATION CONDITIONS

EMERGENCY OPENINGS

1. A utility company may carry out an emergency opening and repair but should contact council as soon as possible after the event.
2. If the restoration subsides within 6 months of the works, the utility company will be required to make good the subsidence, and may be required to pay inspection fees.

RESTORATION CONDITIONS FOR A BITUMEN ROADWAY

1. Conform to drawing on Sheet 1 “Service Trench Road Restoration up to 1.2m Width".
2. When the road surface is asphaltic concrete (“hot mix”) the trench edges will be cut with a concrete saw.
3. Trench widths will kept to a minimum, but with a minimum of 300mm to obtain suitable compaction.
4. All utility mains or services installed across a road (including bored installations) will be a minimum of 600mm (450mm in rock) below the road finished surface level at the line of existing or proposed kerb and gutter.
5. All utility mains or services will be installed at least 300mm below the invert of any open drain.

TEMPORARY RESTORATION - STAGE 1

1. The trench will be backfilled with DGB 20 1.5% stabilised, thoroughly compacted in layers not exceeding 150mm in depth. Where traffic volumes are high, the top 40mm will be “cold mix”. Note: Council approval is required for the use of excavated road base.
2. Where Council is to complete final restoration the contractor is responsible for the restoration for 60 days after Council has been notified that the work is completed. Please notify Council at driveway@lakemac.nsw.gov.au and quote the AM number when the temporary restoration job is completed.

FINAL RESTORATION - STAGE 2

1. The “cold mix” will be milled out to 450mm both sides of the trench and replaced with 50mm of asphalt.
2. Please notify Council at driveway@lakemac.nsw.gov.au and quote the AM number when the job is completed so the warranty period can commence.
3. Where the Permit Holder undertakes the final restoration and it subsides within 6 months from when Council has been notified that the job has been completed, the Permit Holder will be required to make good the subsidence, and may be required pay additional inspection fees.

VERSION	DATE	COMMENTS
01	06/08/2016	ORIGINAL ISSUE

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DRAWING TITLE
SERVICE TRENCH ROAD
RESTORATION UP TO 1.2m WIDTH

SHEET: 2 OF 2

DRAWING No. EGSD-704

VERSION: 01